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Brand Names

First-Pantoprazole, Protonix

Indication Specific Dosing

For the treatment of erosive esophagitis (erosive GERD)

Oral dosage

Adults

40 mg PO once daily for up to 8 weeks. May continue treatment for an additional 8 weeks if healing is incomplete. Controlled studies did not extend beyond 12 months.

Children and Adolescents 5 to 17 years weighing 40 kg or more

40 mg PO once daily for up to 8 weeks.

Children and Adolescents 5 to 17 years weighing 15 to 39 kg

20 mg PO once daily for up to 8 weeks.

Children 1 to 4 years or weighing less than 15 kg†

0.3 to 1.2 mg/kg/dose (Max: 20 mg/dose) PO once daily for up to 8 weeks. Doses of 15 mg/day for children 1 year and 20 mg/day for children 2 to 5 years have been used.

Infants†

1.2 mg/kg/dose PO once daily based on limited data. In a study, 1.2 mg/kg/day PO was administered to infants 1 to 11 months (mean age 5.1 months, n = 128) with GERD symptoms after 2 weeks of conservative treatment. GERD symptom scores decreased significantly from baseline during pantoprazole therapy (p less

than 0.001); treatment was well tolerated with no difference in withdrawal rates or mild-to-moderate adverse events between pantoprazole and placebo groups.

Neonates†

0.6 to 1.2 mg/kg/dose PO once daily based on limited data. Average doses range from 1.25 to 2.5 mg/day.

Intravenous dosage

Adults

40 mg IV injection (over at least 2 minutes) or IV infusion (over 15 minutes) once daily for up to 10 days. Transition to oral therapy as soon as possible.

Children and Adolescents weighing more than 40 kg

40 mg IV infusion over 15 minutes once daily for up to 7 days. Transition to oral therapy as soon as possible.

Children and Adolescents weighing 16 to 40 kg

20 mg IV infusion over 15 minutes once daily for up to 7 days. Transition to oral therapy as soon as possible.

Children and Adolescents weighing 15 kg or less

10 mg IV infusion over 15 minutes once daily for up to 7 days. Transition to oral therapy as soon as possible.

Infants 3 to 11 months weighing 12.5 kg or more

10 mg IV infusion over 15 minutes once daily for up to 7 days. Transition to oral therapy as soon as possible.

Infants 3 to 11 months weighing less than 12.5 kg

0.8 mg/kg IV infusion over 15 minutes once daily for up to 7 days. Transition to oral therapy as soon as possible.

For the treatment of non-erosive gastroesophageal reflux disease (GERD)†

Oral dosage

Adults

40 mg PO once daily for 4 to 8 weeks. May increase the dose to 40 mg PO twice daily in persons with partial response to once daily therapy. Continue maintenance therapy at the lowest effective dose, including on demand or intermittent therapy, in persons who continue to have symptoms after discontinuation.

Children and Adolescents 5 to 17 years weighing 40 kg or more

40 mg PO once daily for up to 8 weeks. Safety beyond 8 weeks has not been established.

Children and Adolescents 5 to 17 years weighing 15 to 39 kg

20 mg PO once daily for up to 8 weeks. Safety beyond 8 weeks has not been established.

Children 1 to 5 years or weighing less than 15 kg

0.6 to 1.2 mg/kg/day PO is suggested from pharmacokinetic data and pediatric reviews; the higher dose is usually prescribed for erosive disease. Doses as low as 0.3 mg/kg/day PO have been studied for symptom reduction of GERD. Some studies for high dosing in erosive esophagitis have allowed for dosing of 15 mg/day PO for patients aged 1 year, and 20 mg/day PO for children aged 2 to 5 years.

Infants

Limited data available. 1.2 mg/kg/day PO is the suggested dose. In 1 study, 1.2 mg/kg/day PO was administered to infants aged 1 to 11 months (mean age 5.1 months, n = 128) with GERD symptoms after 2 weeks of conservative treatment. GERD symptom scores decreased significantly from baseline during pantoprazole therapy ($p < 0.001$); treatment was well tolerated with no difference in withdrawal rates or mild-to-moderate adverse events between pantoprazole and placebo groups. PPIs should not be prescribed as first line therapy for symptomatic GERD in otherwise healthy infants (aged 1 to 11 months); treatment should instead be reserved for use in infants with acid reflux disease diagnosed by endoscopy (e.g., erosive esophagitis) and nonpharmacologic measures such as diet modification and positioning strategies are recommended.

Neonates

Safety and efficacy have not been established. Data are limited. 0.6 to 1.2 mg/kg/day PO (as oral suspension) is a suggested dose range from pharmacokinetic data and pediatric reviews. Average doses were 1.25 mg/day PO (at roughly 0.6 mg/kg/day), and 2.5 mg/day PO (at roughly 1.2 mg/kg/day).

For the treatment of pathological hypersecretion conditions, including Zollinger-Ellison syndrome

Oral dosage

Adults

40 mg PO twice daily, initially. Adjust dose based on clinical response and tolerability. Max: 240 mg/day. Continue treatment for as long as clinically indicated; some individuals with Zollinger-Ellison syndrome have been treated for more than 2 years.

Intravenous dosage

Adults

80 mg IV every 12 hours, initially. Adjust dose based on clinical response and tolerability. Max: 240 mg/day in divided doses every 8 or 12 hours. The safety and efficacy of IV pantoprazole for this indication have not been established beyond 7 days.

For the healing of duodenal ulcer†

Oral dosage

Adults

40 mg PO once daily after the morning meal for 2 to 4 weeks. NOTE: The safe and effective use of pantoprazole for long-term maintenance therapy (e.g., more than 16 weeks) for duodenal ulcer disease has not been established.

For the healing of gastric ulcer†

Oral dosage

Adults

40 mg PO once daily, until healing and symptom relief occurs (e.g., for 2 to 8 weeks), has been evaluated in clinical trials. Healing of gastric ulcers with proton

pump inhibitors generally takes longer than duodenal ulcers and may require 4 to 8 weeks of therapy. The safe and effective use of pantoprazole for long-term maintenance therapy (e.g., more than 16 weeks) of gastric ulcer disease has not been established.

For Helicobacter pylori (H. pylori) eradication†

As part of initial clarithromycin-based therapy in adults without previous macrolide exposure in regions where clarithromycin resistance is less than 15%

Oral dosage

Adults

40 or 80 mg PO twice daily in combination with clarithromycin and either amoxicillin or metronidazole for 14 days.

As part of initial clarithromycin-based therapy in adults with or without potential macrolide exposure or resistance

Oral dosage

Adults

40 mg PO twice daily as part of a combination therapy as a first-line treatment option. Quadruple therapy includes a proton pump inhibitor (PPI) in combination with clarithromycin, amoxicillin, and metronidazole for 10 to 14 days. Hybrid therapy includes amoxicillin plus PPI for 7 days followed by PPI in combination with clarithromycin, amoxicillin, and metronidazole for 7 days. Sequential therapy includes PPI and amoxicillin for 5 to 7 days followed by PPI in combination with clarithromycin and metronidazole for 5 to 7 days.

As part of salvage clarithromycin-based therapy in adults who failed initial bismuth quadruple therapy

Oral dosage

Adults

40 mg PO twice daily in combination with clarithromycin, amoxicillin, and metronidazole for 10 to 14 days. For patients with a penicillin allergy, a PPI is recommended in combination with clarithromycin and metronidazole for 14 days.

In combination with amoxicillin and metronidazole in pediatric patients

Oral dosage

Children and Adolescents weighing 35 kg or more

40 mg PO twice daily in combination with amoxicillin and metronidazole for 14 days. Triple therapy with standard-dose amoxicillin, metronidazole, and a proton pump inhibitor (PPI) is a first-line treatment option for patients infected with H. pylori strains with known susceptibility to metronidazole and resistance to clarithromycin. Triple therapy with high-dose amoxicillin, metronidazole, and a PPI is a first-line treatment option for patients infected with H. pylori strains with dual resistance to clarithromycin and metronidazole or strains with unknown susceptibility.

Children and Adolescents weighing 25 to 34 kg

30 mg PO twice daily in combination with amoxicillin and metronidazole for 14 days. Triple therapy with standard-dose amoxicillin, metronidazole, and a proton pump inhibitor (PPI) is a first-line treatment option for patients infected with H. pylori strains with known susceptibility to metronidazole and resistance to clarithromycin. Triple therapy with high-dose amoxicillin, metronidazole, and a PPI is a first-line treatment option for patients infected with H. pylori strains with dual resistance to clarithromycin and metronidazole or strains with unknown susceptibility.

Children weighing 15 to 24 kg

20 mg PO twice daily in combination with amoxicillin and metronidazole for 14 days. Triple therapy with standard-dose amoxicillin, metronidazole, and a proton pump inhibitor (PPI) is a first-line treatment option for patients infected with H. pylori strains with known susceptibility to metronidazole and resistance to clarithromycin. Triple therapy with high-dose amoxicillin, metronidazole, and a PPI is a first-line treatment option for patients infected with H. pylori strains with dual resistance to clarithromycin and metronidazole or strains with unknown susceptibility.

In combination with amoxicillin and clarithromycin in pediatric patients

Oral dosage

Children and Adolescents weighing 35 kg or more

40 mg PO twice daily in combination with amoxicillin and clarithromycin for 14 days. Triple therapy with standard-dose amoxicillin, clarithromycin, and a proton pump inhibitor is a first-line treatment option for patients infected with fully susceptible H. pylori strains or strains susceptible to clarithromycin but resistant to metronidazole. In cases of penicillin allergy, use metronidazole in place of amoxicillin for patients infected with fully susceptible strains.

Children and Adolescents weighing 25 to 34 kg

30 mg PO twice daily in combination with amoxicillin and clarithromycin for 14 days. Triple therapy with standard-dose amoxicillin, clarithromycin, and a proton pump inhibitor is a first-line treatment option for patients infected with fully susceptible H. pylori strains or strains susceptible to clarithromycin but resistant to metronidazole. In cases of penicillin allergy, use metronidazole in place of amoxicillin for patients infected with fully susceptible strains.

Children weighing 15 to 24 kg

20 mg PO twice daily in combination with amoxicillin and clarithromycin for 14 days. Triple therapy with standard-dose amoxicillin, clarithromycin, and a proton pump inhibitor is a first-line treatment option for patients infected with fully susceptible H. pylori strains or strains susceptible to clarithromycin but resistant to metronidazole. In cases of penicillin allergy, use metronidazole in place of amoxicillin for patients infected with fully susceptible strains.

As part of a sequential therapy regimen in pediatric patients

Oral dosage

Children and Adolescents weighing 35 kg or more

40 mg PO twice daily for 10 days. Use in combination with amoxicillin for days 1 through 5, and then clarithromycin and metronidazole for days 6 through 10. Sequential therapy is a first-line treatment option for patients infected with fully susceptible H. pylori strains. Sequential therapy is not recommended if susceptibility testing is unavailable.

Children and Adolescents weighing 25 to 34 kg

30 mg PO twice daily for 10 days. Use in combination with amoxicillin for days 1

through 5, and then clarithromycin and metronidazole for days 6 through 10. Sequential therapy is a first-line treatment option for patients infected with fully susceptible H. pylori strains. Sequential therapy is not recommended if susceptibility testing is unavailable.

Children weighing 15 to 24 kg

20 mg PO twice daily for 10 days. Use in combination with amoxicillin for days 1 through 5, and then clarithromycin and metronidazole for days 6 through 10. Sequential therapy is a first-line treatment option for patients infected with fully susceptible H. pylori strains. Sequential therapy is not recommended if susceptibility testing is unavailable.

As part of a quadruple therapy regimen in pediatric patients

Oral dosage

Children and Adolescents weighing 35 kg or more

40 mg PO twice daily in combination with amoxicillin, metronidazole, and clarithromycin for 14 days. Concomitant quadruple therapy with amoxicillin, metronidazole, clarithromycin, and a proton pump inhibitor is a first-line treatment option for patients infected with H. pylori strains with dual resistance to clarithromycin and metronidazole or strains with unknown susceptibility

Children and Adolescents weighing 25 to 34 kg

30 mg PO twice daily in combination with amoxicillin, metronidazole, and clarithromycin for 14 days. Concomitant quadruple therapy with amoxicillin, metronidazole, clarithromycin, and a proton pump inhibitor is a first-line treatment option for patients infected with H. pylori strains with dual resistance to clarithromycin and metronidazole or strains with unknown susceptibility.

Children weighing 15 to 24 kg

20 mg PO twice daily in combination with amoxicillin, metronidazole, and clarithromycin for 14 days. Concomitant quadruple therapy with amoxicillin, metronidazole, clarithromycin, and a proton pump inhibitor is a first-line treatment option for patients infected with H. pylori strains with dual resistance to clarithromycin and metronidazole or strains with unknown susceptibility.

As part of levofloxacin-based initial therapy in adults

Oral dosage

Adults

40 or 80 mg PO twice daily as part of combination therapy as a first-line treatment option. Triple therapy includes pantoprazole 40 mg PO twice daily in combination with levofloxacin and amoxicillin for 10 to 14 days. Sequential therapy includes pantoprazole 40 or 80 mg PO twice daily in combination with amoxicillin for 5 to 7 days followed by pantoprazole 40 mg PO twice daily in combination with levofloxacin and a nitroimidazole for 5 to 7 days. Quadruple therapy includes pantoprazole 80 mg PO once daily in combination with levofloxacin, nitazoxanide, and doxycycline for 7 to 10 days.

As part of levofloxacin-based salvage therapy in adults

Oral dosage

Adults

40 mg PO twice daily in combination with levofloxacin and amoxicillin for 14 days. Guidelines recommend this triple therapy in patients who have failed clarithromycin-triple or bismuth-quadruple initial therapies and without previous quinolone exposure. Levofloxacin in combination with metronidazole and a PPI for 14 days could be considered for patients with a penicillin allergy who have failed prior bismuth quadruple therapy.

As part of bismuth-based initial therapy in adults

Oral dosage

Adults

40 mg PO twice daily in combination with bismuth subcitrate or subsalicylate, metronidazole, and tetracycline for 10 to 14 days is recommended as a first-line treatment option, particularly in patients with any previous macrolide exposure or a penicillin allergy.

As part of rifabutin-based salvage therapy in adults

Oral dosage

Adults

40 mg PO twice daily in combination with rifabutin and amoxicillin for 10 days.

As part of high-dose dual salvage therapy in adults

Oral dosage

Adults

40 or 80 mg PO 3 or 4 times daily in combination with high-dose amoxicillin for 14 days. A high-dose proton pump inhibitor in combination with metronidazole may be considered in patients with prior quinolone exposure and a penicillin allergy who have failed initial bismuth quadruple therapy.

As part of bismuth-based quadruple salvage therapy in adults

Oral dosage

Adults

40 mg PO twice daily in combination with bismuth subcitrate or subsalicylate, tetracycline, and metronidazole for 14 days is recommended particularly in patients failing clarithromycin triple therapy. A subsequent repeat course of bismuth quadruple therapy may be considered after failed prior bismuth quadruple therapy.

For NSAID-induced ulcer prophylaxis†

Oral dosage

Adults

40 mg PO once daily has been studied. Patients (receiving chronic NSAID therapy for rheumatoid conditions) had either normal GI mucosa or endoscopically confirmed mucosal changes or ulcers (Lanza classification grade 0, 1, or 2) at baseline. Patients were randomized to either pantoprazole 40 mg PO once daily or placebo. The primary end point of the study was the incidence of gastric or duodenal ulcers after 4 and 12 weeks of treatment. In patients with normal GI mucosa (Lanza grade 0) at baseline, 82% (CI, 70%—94%) of those receiving pantoprazole were in remission vs. 55% (CI, 33%—77%) with placebo at 12 weeks, a statistically significant difference. In all patients (Lanza grade 0, 1, and 2) with GI lesions at baseline, remission was achieved in 72% of those receiving pantoprazole vs. 59% of the placebo group.

For stress gastritis prophylaxis† in critically-ill patients

Intravenous dosage

Adults

40 mg IV once daily; some studies suggest more frequent dosing may be more effective. The optimal dose and duration of treatment has not been established. A usual dose of 40 mg/day led to an upper GI bleed in 3 of 95 mechanically ventilated patients in 1 retrospective study; thus, higher pantoprazole doses (e.g., increased doses and more frequent intervals of dosing) may be needed to prevent stress ulceration. In another study, after 2 days of pantoprazole, the mean time that gastric aspirates were pH of 4 or higher was 67.5% +/- 34.4% with 40 mg/day and was 85.7% +/- 21.5% with 80 mg every 8 hours. For those patients with risk factors qualifying them for stress ulcer prophylaxis, the H2 antagonists, cytoprotective agents, and some PPIs are adequate choices. Optimum duration is unclear, but it is reasonable to provide therapy while risk factors are present, the patient is in a critical care unit, or for at least 1 week after onset of critical illness.

Infants, Children, and Adolescents

Data are limited; optimal dosing and safety have not been established. 40 mg/1.73 m2/day IV infusion once daily was the initial dose used in patients older than 1 month in a small pharmacokinetic/pharmacodynamic study (n = 20; age range: 10 days to 16.4 years). The initial dosage regimen was extrapolated from the recommended adult dose scaled to body surface area. The median effective dose for all patients was 41.8 mg/1.73 m2/day (range: 19.9 to 80 mg) or approximately 1.1 mg/kg/day (range: 0.5 to 2 mg/kg/day). Max: 80 mg/1.73 m2/day (approximately 2 mg/kg/day) was allowed for patients with inadequate gastric acid suppression. The authors concluded the data best predicted suggested dosing for pediatric patients aged 1 month to 5 years. More studies are needed to confirm optimal dosing regimens. Very few adolescents were included in the study sample.

Neonates

Data are very limited in neonates; optimal dosing and safety have not been established. 20 mg/1.73 m2/day IV infusion once daily (approximately 0.5 mg/kg/day) was the initial neonatal dose used in a small pharmacokinetic/pharmacodynamic study (n = 20; age range: 10 days to 16.4 years). The initial dosage regimen was extrapolated from the recommended adult dose scaled to body surface area. The median effective dose for all patients was 41.8 mg/1.73 m2/day (range: 19.9 to 80 mg) or approximately 1.1 mg/kg/day (range: 0.5 to 2 mg/kg/day). Max: 80 mg/1.73 m2/day (approximately 2 mg/kg/day) was allowed for patients with inadequate gastric acid suppression. The authors concluded the data best predicted suggested dosing for pediatric

patients aged 1 month to 5 years. More studies are needed to confirm optimal dosing regimens. Very few neonates were included in the study sample.

Oral dosage

Adults

40 mg PO once daily; some studies suggest more frequent dosing may be more effective. The optimal dose and duration of treatment has not been established. A usual dose of 40 mg/day led to an upper GI bleed in 3 of 95 mechanically ventilated patients in 1 retrospective study; thus, higher pantoprazole doses (e.g., increased doses and more frequent intervals of dosing) may be needed to prevent stress ulceration. In another study, after 2 days of pantoprazole, the mean time that gastric aspirates were pH of 4 or higher was 67.5% +/- 34.4% with 40 mg/day and was 85.7% +/- 21.5% with 80 mg every 8 hours. For those patients with risk factors qualifying them for stress ulcer prophylaxis, the H2 antagonists, cytoprotective agents, and some PPIs are adequate choices. Optimum duration is unclear, but it is reasonable to provide therapy while risk factors are present, the patient is in a critical care unit, or for at least 1 week after onset of critical illness.

For the treatment of eosinophilic esophagitis (EoE)†

Oral dosage

Adults

A dosage range of 20 to 40 mg PO twice daily 30 to 60 minutes before meals has been suggested; treat for up to 8 weeks and continue until the time of the follow-up endoscopy and biopsy. The guidelines support the use of PPI therapy for EoE based on reports of reductions in histologic features of disease from 42% in observational studies.

For upper GI rebleeding prophylaxis† after therapeutic endoscopy

For upper GI rebleeding prophylaxis† after therapeutic endoscopy as short-term high-dose proton pump inhibitor therapy

Oral dosage

Adults

80 mg PO as a single dose, followed by 40 mg PO 2 to 4 times daily for 72 hours after successful endoscopic hemostasis.

Intravenous dosage

Adults

80 mg IV as a single dose, followed by 40 mg IV 2 to 4 times daily for 72 hours after successful endoscopic hemostasis.

Continuous Intravenous Infusion dosage

Adults

80 mg IV as a single dose, followed by 8 mg/hour continuous IV infusion for 72 hours after successful endoscopic hemostasis.

For upper GI rebleeding prophylaxis† after therapeutic endoscopy in high-risk persons with upper GI bleeding due to ulcers who received endoscopic therapy followed by short-term high-dose proton pump inhibitor therapy

Oral dosage

Adults

40 mg PO twice daily from days 4 to 14 after the index endoscopy and initial 3 days of high-dose proton pump inhibitor therapy.

For the treatment of short bowel syndrome†

Oral dosage

Adults

20 to 40 mg PO twice daily for at least 6 months.

Intravenous dosage

Adults

20 to 40 mg IV twice daily for at least 6 months.

For the short-term treatment of frequent dyspepsia† or pyrosis (heartburn)† that occurs 2 or more times per week

Oral dosage

Adults

20 to 40 mg PO once daily for up to 14 days. Full relief may take 1 to 4 days. Reassess if frequent heartburn returns soon after the 14-day treatment regimen.

Contraindications And Precaution

Drug Interactions

The coadministration of certain medications may lead to harm and require avoidance or therapy modification; review all drug interactions prior to concomitant use of other medications.

Hypersensitivity

This medication is contraindicated in patients with a history of hypersensitivity to it or any of its components. Pantoprazole is also contraindicated in individuals with a history of hypersensitivity to other substituted benzimidazoles [e.g., other proton pump inhibitors (PPIs)]. Before starting therapy with pantoprazole, carefully inquire about previous hypersensitivity reactions to PPIs.

General Information

In adults, symptomatic response to therapy with a proton pump inhibitor (PPI) does not preclude the presence of gastric malignancy. Consider additional follow-up and diagnostic testing in adults who have a suboptimal response or an early symptomatic relapse after completing treatment with a PPI. In older adult patients, also consider an endoscopy.

hypocalcemia, hypokalemia, hypomagnesemia, hypoparathyroidism

Use pantoprazole with caution in people with a pre-existing risk of hypocalcemia (e.g., hypoparathyroidism), hypokalemia, or hypomagnesemia; consider monitoring magnesium and calcium concentrations prior to initiating therapy and periodically while on treatment in at-risk patients. Supplement with magnesium and/or calcium as needed and consider discontinuing proton pump inhibitor (PPI) therapy if hypomagnesemia or hypocalcemia is refractory to treatment.

zinc deficiency

The intravenous formulation of pantoprazole contains edetate disodium (the salt form of EDTA), a potent chelator of metal ions including zinc. Clinicians should consider zinc supplementation in patients treated with intravenous pantoprazole who are prone to zinc deficiency.

increased risk for osteoporosis

Use proton pump inhibitors (PPIs) with caution in people with an increased risk for osteoporosis. Several published observational studies suggest that PPI therapy may be associated with an increased risk for osteoporosis-related fractures of the hip, wrist, or spine. The risk of fracture was increased in patients who received high-dose, defined as multiple daily doses, and long-term PPI therapy (a year or longer). Use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated. People at risk for osteoporosis-related fractures should be managed according to established treatment guidelines.

cutaneous lupus erythematosus, systemic lupus erythematosus

Cutaneous lupus erythematosus (CLE) and systemic lupus erythematosus (SLE) have been reported in people taking proton pump inhibitors (PPIs), including pantoprazole. These events have occurred as both new onset and an exacerbation of pre-existing autoimmune disease. Avoid use of PPIs for longer than medically indicated in these patients.

geriatric

According to the Beers Criteria, proton pump inhibitors (PPIs) are considered potentially inappropriate medications (PIMs) for use in geriatric adults due to the risk of Clostridium difficile infection, pneumonia, GI malignancies, and bone loss/fractures. Avoid scheduled use for more than 8 weeks except for high-risk adults (e.g., oral corticosteroids or chronic NSAID use), erosive esophagitis, Barrett's esophagitis, pathological hypersecretory condition, or a demonstrated need for maintenance treatment (e.g., due to failure of a drug discontinuation trial or inadequate response to H2-receptor blockers).

pregnancy

Available observational data do not show a clear association between pantoprazole use during pregnancy and major malformations or adverse outcomes. However, methodological limitations prevent definitive exclusion of drug-related risks. A prospective study by the European Network of Teratology Information Services evaluated 53 pregnant individuals with a median daily dose of 40 mg pantoprazole, compared to 868 controls not using proton pump inhibitors (PPIs). No difference was found in major malformation rates (RR = 0.55; 95% CI 0.08 to 3.95). Similarly, a population-based retrospective cohort study in Denmark (1996 to 2008) of 549 live births exposed to pantoprazole in the first trimester showed no significant increase in major birth defects. Supporting these findings, a meta-analysis of 1,530 pregnant

individuals exposed to PPIs during the first trimester versus 133,410 unexposed showed no significant increase in major malformations (OR = 1.12; 95% CI 0.86 to 1.45) or spontaneous abortion (OR = 1.29; 95% CI 0.84 to 1.97). In animal studies, reproductive toxicity was not observed. Specifically, in rats and rabbits, intravenous pantoprazole administered during organogenesis at doses 4 to 6 times the human dose based on body surface area (BSA) showed no evidence of fetal harm. Oral doses at 16 to 88 times the human dose based on BSA also did not produce fetal toxicity. However, in rat pre- and postnatal development studies, oral pantoprazole at doses as low as approximately equal to the human dose based on BSA, and given from gestation through lactation, caused changes in offspring's bone morphology. Guidelines recommend a trial of lifestyle modifications as first-line therapy for heartburn and gastroesophageal reflux disease (GERD) during pregnancy, followed by antacids if lifestyle adjustments are ineffective. For ongoing symptoms, histamine type 2-receptor antagonists (H2RAs) can be used. PPIs should be reserved for pregnant patients who fail H2RA therapy.

breast-feeding

Pantoprazole is compatible with breast-feeding. Pantoprazole is present in human milk in small amounts, with a relative infant dose between 0.14% and 1%. Pantoprazole exposure via human milk is unlikely to cause side effects in the breast-fed child. There are no data on the effect of pantoprazole on milk production. Proton pump inhibitors may not be the preferred first-line agent. If heartburn or gastroesophageal reflux disease symptoms persist after delivery, antacids and sucralfate are also compatible with breast-feeding due to minimal presence in human milk. Histamine type 2-receptor antagonists like cimetidine and famotidine are considered compatible with breast-feeding and may be used if symptoms persist despite antacid use.

Pregnancy And Lactation

Available observational data do not show a clear association between pantoprazole use during pregnancy and major malformations or adverse outcomes. However, methodological limitations prevent definitive exclusion of drug-related risks. A prospective study by the European Network of Teratology Information Services evaluated 53 pregnant individuals with a median daily dose of 40 mg pantoprazole, compared to 868 controls not using proton pump inhibitors (PPIs). No difference was found in major malformation rates (RR = 0.55; 95% CI 0.08 to 3.95). Similarly, a population-based retrospective cohort study in Denmark (1996 to 2008) of 549 live births exposed to pantoprazole in the first trimester showed no significant increase in major birth defects. Supporting these findings, a meta-analysis of 1,530 pregnant individuals exposed to PPIs during the first trimester versus 133,410 unexposed showed

no significant increase in major malformations (OR = 1.12; 95% CI 0.86 to 1.45) or spontaneous abortion (OR = 1.29; 95% CI 0.84 to 1.97). In animal studies, reproductive toxicity was not observed. Specifically, in rats and rabbits, intravenous pantoprazole administered during organogenesis at doses 4 to 6 times the human dose based on body surface area (BSA) showed no evidence of fetal harm. Oral doses at 16 to 88 times the human dose based on BSA also did not produce fetal toxicity. However, in rat pre- and postnatal development studies, oral pantoprazole at doses as low as approximately equal to the human dose based on BSA, and given from gestation through lactation, caused changes in offspring's bone morphology. Guidelines recommend a trial of lifestyle modifications as first-line therapy for heartburn and gastroesophageal reflux disease (GERD) during pregnancy, followed by antacids if lifestyle adjustments are ineffective. For ongoing symptoms, histamine type 2-receptor antagonists (H2RAs) can be used. PPIs should be reserved for pregnant patients who fail H2RA therapy.

Interactions

Acalabrutinib: (Major) Avoid the concomitant use of acalabrutinib capsules and proton pump inhibitors (PPI), such as pantoprazole; decreased acalabrutinib exposure may occur resulting in decreased acalabrutinib effectiveness. Consider using the acalabrutinib tablet formlation or use an antacid or H2-blocker if acid suppression therapy is needed. Separate the administration of acalabrutinib capsules and antacids by at least 2 hours; give acalabrutinib capsules 2 hours before a H2-blocker. Acalabrutinib capsule solubility decreases with increasing pH values. The AUC of acalabrutinib was decreased by 43% when acalabrutinib capsules were coadministered with another PPI for 5 days.
Albuterol; Budesonide: (Minor) Enteric-coated budesonide granules dissolve at a pH greater than 5.5. Concomitant use of budesonide oral capsules and drugs that increase gastric pH levels can cause the coating of the granules to dissolve prematurely, possibly affecting release properties and absorption of the drug in the duodenum.
Alendronate: (Moderate) Proton pump inhibitors (PPIs) are widely used and are frequently coadministered in users of oral bisphosphonates. A national register-based, open cohort study of 38,088 elderly patients suggests that those who use proton pump inhibitors in conjunction with alendronate have a dose-dependent loss of protection against hip fracture. While causality was not investigated, the dose-response relationship noted during the study suggested that PPIs may reduce oral alendronate efficacy, perhaps through an effect on absorption or other mechanism, and therefore PPIs may not be optimal agents to control gastrointestinal complaints. It is not yet clear if all bisphosphonates would exhibit a loss of efficacy when PPIs are coadministered, but the results suggest that the interaction may occur across the class.
Alendronate; Cholecalciferol: (Moderate) Proton pump inhibitors (PPIs) are widely used and are frequently coadministered in users of oral bisphosphonates. A national register-

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Aliskiren; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
aMILoride; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
amLODIPine; Valsartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Amphetamine: (Moderate) Use amphetamine; dextroamphetamine and proton pump inhibitors concomitantly with caution. Gastrointestinal alkalinizing agents may increase exposure to amphetamine; dextroamphetamine and exacerbate its actions.
Amphetamine; Dextroamphetamine Salts: (Moderate) Use amphetamine; dextroamphetamine and proton pump inhibitors concomitantly with caution. Gastrointestinal alkalinizing agents may increase exposure to amphetamine; dextroamphetamine and exacerbate its actions.
Amphetamine; Dextroamphetamine: (Moderate) Use amphetamine; dextroamphetamine and proton pump inhibitors concomitantly with caution. Gastrointestinal alkalinizing agents may increase exposure to amphetamine; dextroamphetamine and exacerbate its actions.
Ampicillin: (Major) Proton pump inhibitors (PPIs) have long-lasting effects on the secretion of gastric acid. For enteral ampicillin, whose bioavailability is influenced by gastric pH, the concomitant administration of PPIs can exert a significant effect on ampicillin absorption.
Ampicillin; Sulbactam: (Major) Proton pump inhibitors (PPIs) have long-lasting effects on the secretion of gastric acid. For enteral ampicillin, whose bioavailability is influenced by gastric pH, the concomitant administration of PPIs can exert a significant effect on ampicillin absorption.
Atazanavir: (Contraindicated) Coadministration of proton pump inhibitors (PPIs) with atazanavir in treatment-experienced patients is contraindicated. PPIs can be used with atazanavir in treatment-naive patients under specific administration restrictions. In treatment-naive patients >= 40 kg, the PPI dose should not exceed the equivalent of

omeprazole 20 mg/day, and the PPI must be administered 12 hours before atazanavir and ritonavir; use the dosage regimen of atazanavir 300 mg boosted with ritonavir 100 mg given once daily with food. While data are insufficient to recommend atazanavir dosing in children < 40 kg receiving concomitant PPIs, the same recommendations regarding timing and maximum doses of concomitant PPIs should be followed. Closely monitor patients for antiretroviral therapeutic failure and resistance development during treatment with a PPI. A randomized, open-label, multiple-dose drug interaction study of atazanavir (300 mg) with ritonavir (100 mg) coadministered with omeprazole 40 mg found a reduction in atazanavir AUC and Cmin of 76% and 78%, respectively. Additionally, after multiple doses of omeprazole (40 mg/day) and atazanavir (400 mg/day, 2 hours after omeprazole) without ritonavir, the AUC of atazanavir was decreased by 94%, Cmax by 96%, and Cmin by 95%.
Atazanavir; Cobicistat: (Contraindicated) Coadministration of proton pump inhibitors (PPIs) with atazanavir in treatment-experienced patients is contraindicated. PPIs can be used with atazanavir in treatment-naive patients under specific administration restrictions. In treatment-naive patients >= 40 kg, the PPI dose should not exceed the equivalent of omeprazole 20 mg/day, and the PPI must be administered 12 hours before atazanavir and ritonavir; use the dosage regimen of atazanavir 300 mg boosted with ritonavir 100 mg given once daily with food. While data are insufficient to recommend atazanavir dosing in children < 40 kg receiving concomitant PPIs, the same recommendations regarding timing and maximum doses of concomitant PPIs should be followed. Closely monitor patients for antiretroviral therapeutic failure and resistance development during treatment with a PPI. A randomized, open-label, multiple-dose drug interaction study of atazanavir (300 mg) with ritonavir (100 mg) coadministered with omeprazole 40 mg found a reduction in atazanavir AUC and Cmin of 76% and 78%, respectively. Additionally, after multiple doses of omeprazole (40 mg/day) and atazanavir (400 mg/day, 2 hours after omeprazole) without ritonavir, the AUC of atazanavir was decreased by 94%, Cmax by 96%, and Cmin by 95%.
Atenolol; Chlorthalidone: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Avutometinib; Defactinib: (Major) Avoid concomitant use of defactinib and proton pump inhibitors (PPIs). Concurrent use interferes with defactinib absorption which may decrease defactinib exposure and efficacy. Concomitant use with a PPI reduced defactinib overall exposure by 79% and reduced the exposure of the M4 active metabolite by 83%.
Azilsartan; Chlorthalidone: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Belumosudil: (Major) Increase the dosage of belumosudil to 200 mg PO twice daily when

coadministered with a proton pump inhibitor (PPI). Concomitant use may result in decreased belumosudil exposure and reduced belumosudil efficacy. Coadministration with other PPIs has decreased belumosudil exposure by 47% to 80% in healthy subjects.
Benazepril; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Bisacodyl: (Minor) The concomitant use of bisacodyl oral tablets with drugs that raise gastric pH like proton pump inhibitors can cause the enteric coating of the bisacodyl tablets to dissolve prematurely, leading to possible gastric irritation or dyspepsia. When taking bisacodyl tablets, it is advisable to avoid PPIs within 1 hour before or after the bisacodyl dosage.
Bisoprolol; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Bosutinib: (Major) Bosutinib displays pH-dependent aqueous solubility; therefore, concomitant use of bosutinib and proton-pump inhibitors, such as pantoprazole, may result in decreased plasma exposure of bosutinib. Consider using a short-acting antacid or H2 blocker if acid suppression therapy is needed; separate the administration of bosutinib and antacids or H2-blockers by more than 2 hours.
Budesonide: (Minor) Enteric-coated budesonide granules dissolve at a pH greater than 5.5. Concomitant use of budesonide oral capsules and drugs that increase gastric pH levels can cause the coating of the granules to dissolve prematurely, possibly affecting release properties and absorption of the drug in the duodenum.
Budesonide; Formoterol: (Minor) Enteric-coated budesonide granules dissolve at a pH greater than 5.5. Concomitant use of budesonide oral capsules and drugs that increase gastric pH levels can cause the coating of the granules to dissolve prematurely, possibly affecting release properties and absorption of the drug in the duodenum.
Budesonide; Glycopyrrolate; Formoterol: (Minor) Enteric-coated budesonide granules dissolve at a pH greater than 5.5. Concomitant use of budesonide oral capsules and drugs that increase gastric pH levels can cause the coating of the granules to dissolve prematurely, possibly affecting release properties and absorption of the drug in the duodenum.
Bumetanide: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and loop diuretic use due to risk for hypomagnesemia.
Cabotegravir; Rilpivirine: (Contraindicated) Concurrent use of proton pump inhibitors and rilpivirine is contraindicated; when these drugs are coadministered, there is a potential for treatment failure and/or the development of rilpivirine or NNRTI resistance. Proton pump inhibitors inhibit secretion of gastric acid by proton pumps thereby increasing the gastric pH; for optimal absorption, rilpivirine requires an acidic

environment. Coadministration of a proton pump inhibitor and rilpivirine may result in decreased rilpivirine absorption/serum concentrations, which could cause impaired virologic response to rilpivirine.
Candesartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Captopril; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Cefpodoxime: (Moderate) Cefpodoxime proxetil requires a low gastric pH for dissolution; therefore, concurrent administration with medications that increase gastric pH, such as proton pump inhibitors (PPIs) may decrease the bioavailability of cefpodoxime. When cefpodoxime was administered with high doses of antacids and H2-blockers, peak plasma concentrations were reduced by 24% and 42% and the extent of absorption was reduced by 27% and 32%, respectively. The rate of absorption is not affected.
Cefuroxime: (Major) Avoid the concomitant use of proton pump inhibitors (PPIs) and cefuroxime. Drugs that reduce gastric acidity, such as PPIs, can interfere with the oral absorption of cefuroxime axetil and may result in reduced antibiotic efficacy.
Chlorothiazide: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Chlorthalidone: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Cysteamine: (Major) Monitor white blood cell (WBC) cystine concentration closely when administering delayed-release cysteamine (Procysbi) with proton pump inhibitors (PPIs). Drugs that increase the gastric pH may cause the premature release of cysteamine from delayed-release capsules, leading to an increase in WBC cystine concentration. Concomitant administration of omeprazole 20 mg did not alter the pharmacokinetics of delayed-release cysteamine when administered with orange juice; however, the effect of omeprazole on the pharmacokinetics of delayed-release cysteamine when administered with water have not been studied.
Dacomitinib: (Major) Avoid coadministration of pantoprazole with dacomitinib due to decreased plasma concentrations of dacomitinib which may impact efficacy. Coadministration with another proton pump inhibitor decreased the dacomitinib Cmax and AUC by 51% and 39%, respectively.
Dasatinib: (Major) Avoid the concomitant use of proton pump inhibitors (PPIs) with dasatinib film-coated oral tablets, such as Sprycel. Consider using an alternative dasatinib dosage form, such as Phyrago, or antacids. Separate the administration of all

dasatinib oral tablet dosage forms and antacids by at least 2 hours. PPIs alter gastric pH and interfere with the absorption of some dasatinib dosage forms which may reduce dasatinib efficacy. The use of a PPI with Sprycel reduced dasatinib overall exposure by 43%.
Desogestrel: Ethinyl Estradiol; Ferrous Bisglycinate: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Dextroamphetamine: (Moderate) Use amphetamine; dextroamphetamine and proton pump inhibitors concomitantly with caution. Gastrointestinal alkalinizing agents may increase exposure to amphetamine; dextroamphetamine and exacerbate its actions.
Digoxin: (Moderate) Monitor serum magnesium concentrations periodically in patients taking pantoprazole and digoxin concomitantly. Patients who develop hypomagnesemia may require pantoprazole discontinuation in addition to magnesium replacement. Proton pump inhibitors (PPIs) have been associated with hypomagnesemia which may lead to irregular heartbeat and increase the likelihood of serious cardiac arrhythmias. Additionally, monitor patients with digoxin serum concentrations at the upper end of the therapeutic range for potential increases in serum digoxin concentrations when coadministered with pantoprazole. Although pantoprazole has not been shown to increase digoxin bioavailability, other PPIs have slightly increased digoxin levels due to the long-lasting effect of the PPIs on gastric acid secretion, which affects the absorption of digoxin.
Dolutegravir; Rilpivirine: (Contraindicated) Concurrent use of proton pump inhibitors and rilpivirine is contraindicated; when these drugs are coadministered, there is a potential for treatment failure and/or the development of rilpivirine or NNRTI resistance. Proton pump inhibitors inhibit secretion of gastric acid by proton pumps thereby increasing the gastric pH; for optimal absorption, rilpivirine requires an acidic environment. Coadministration of a proton pump inhibitor and rilpivirine may result in decreased rilpivirine absorption/serum concentrations, which could cause impaired virologic response to rilpivirine.
Emtricitabine; Rilpivirine; Tenofovir alafenamide: (Contraindicated) Concurrent use of proton pump inhibitors and rilpivirine is contraindicated; when these drugs are coadministered, there is a potential for treatment failure and/or the development of rilpivirine or NNRTI resistance. Proton pump inhibitors inhibit secretion of gastric acid by proton pumps thereby increasing the gastric pH; for optimal absorption, rilpivirine requires an acidic environment. Coadministration of a proton pump inhibitor and rilpivirine may result in decreased rilpivirine absorption/serum concentrations, which

could cause impaired virologic response to rilpivirine.
Emtricitabine; Rilpivirine; Tenofovir Disoproxil Fumarate: (Contraindicated) Concurrent use of proton pump inhibitors and rilpivirine is contraindicated; when these drugs are coadministered, there is a potential for treatment failure and/or the development of rilpivirine or NNRTI resistance. Proton pump inhibitors inhibit secretion of gastric acid by proton pumps thereby increasing the gastric pH; for optimal absorption, rilpivirine requires an acidic environment. Coadministration of a proton pump inhibitor and rilpivirine may result in decreased rilpivirine absorption/serum concentrations, which could cause impaired virologic response to rilpivirine.
Enalapril; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Erlotinib: (Major) Avoid coadministration of erlotinib with pantoprazole if possible due to decreases in erlotinib plasma concentrations. Erlotinib solubility is pH dependent and solubility decreases as pH increases. Coadministration of erlotinib with medications that increase the pH of the upper gastrointestinal tract may decrease the absorption of erlotinib. Separation of doses may not eliminate the interaction since proton pump inhibitors affect the pH of the upper GI tract for an extended period of time. Increasing the dose of erlotinib is also not likely to compensate for the loss of exposure. Coadministration with another proton pump inhibitor decreased erlotinib exposure by 46% and the erlotinib Cmax by 61%.
Ethacrynic Acid: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and loop diuretic use due to risk for hypomagnesemia.
Ferric Maltol: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Fosinopril; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Furosemide: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and loop diuretic use due to risk for hypomagnesemia.
Gefitinib: (Moderate) Avoid coadministration of pantoprazole with gefitinib if possible due to decreased exposure to gefitinib, which may lead to reduced efficacy. If concomitant use is unavoidable, take gefitinib 12 hours after the last dose or 12 hours

before the next dose of pantoprazole. Gefitinib exposure is affected by gastric pH. Coadministration with another drug to maintain gastric pH above 5 decreased gefitinib exposure by 47%.
hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
hydroCHLOROthiazide, HCTZ; Moexipril: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Infigratinib: (Major) Avoid coadministration of infigratinib and gastric acid-reducing agents, such as proton pump inhibitors (PPIs). Coadministration may decrease infigratinib exposure resulting in decreased efficacy. If necessary, infigratinib may be administered two hours before or ten hours after an H2-receptor antagonist or two hours before or after a locally acting antacid. Coadministration with a PPI decreased infigratinib exposure by 45%.
Irbesartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Iron Salts: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Iron Salts: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Iron: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Itraconazole: (Moderate) When administering proton pump inhibitors with the 100 mg

itraconazole capsule and 200 mg itraconazole tablet formulations, systemic exposure to itraconazole is decreased. Conversely, exposure to itraconazole is increased when proton pump inhibitors are administered with the 65 mg itraconazole capsule. Administer proton pump inhibitors at least 2 hours before or 2 hours after the 100 mg capsule or 200 mg tablet. Monitor for increased itraconazole-related adverse effects if proton pump inhibitors are administered with itraconazole 65 mg capsules.
Ketoconazole: (Major) Avoid use of proton pump inhibitors (PPIs) with ketoconazole. Medications that increase gastric pH may impair oral ketoconazole absorption.
Ledipasvir; Sofosbuvir: (Major) Solubility of ledipasvir decreases as gastric pH increases; thus, coadministration of ledipasvir; sofosbuvir with proton pump inhibitors (PPIs) may result in lower ledipasvir plasma concentrations. Ledipasvir can be administered with PPIs if given simultaneously under fasting conditions. The PPI dose should not exceed a dose that is comparable to omeprazole 20 mg/day.
Letermovir: (Moderate) Monitor for reduced pantoprazole efficacy and adjust the dose of pantoprazole if needed during concurrent use of letermovir. Coadministration may result in a clinically relevant decrease in the plasma concentration of pantoprazole. Pantoprazole is a substrate of CYP2C19. Letermovir is a CYP2C19 inducer.
Levoketoconazole: (Major) Avoid use of proton pump inhibitors (PPIs) with ketoconazole. Medications that increase gastric pH may impair oral ketoconazole absorption.
Levonorgestrel; Ethinyl Estradiol; Ferrous Bisglycinate: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Levonorgestrel; Ethinyl Estradiol; Ferrous Fumarate: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Levothyroxine: (Moderate) Proton pump inhibitors (PPIs) may reduce the oral absorption of thyroid hormones and thus reduce efficacy; monitor for altered clinical response to thyroid hormone therapy if concomitant use is necessary. Alternatively, an oral liquid levothyroxine dosage form may be considered. Gastric acidity is an essential requirement for adequate absorption of levothyroxine tablets and capsules and other thyroid hormones. Gastric acidity may be less essential for the absorption of oral liquid dosage forms of levothyroxine; PPIs have been observed to have a minimal effect on the

bioavailability of oral liquid levothyroxine.
Levothyroxine; Liothyronine (Porcine): (Moderate) Proton pump inhibitors (PPIs) may reduce the oral absorption of thyroid hormones and thus reduce efficacy; monitor for altered clinical response to thyroid hormone therapy if concomitant use is necessary. Alternatively, an oral liquid levothyroxine dosage form may be considered. Gastric acidity is an essential requirement for adequate absorption of levothyroxine tablets and capsules and other thyroid hormones. Gastric acidity may be less essential for the absorption of oral liquid dosage forms of levothyroxine; PPIs have been observed to have a minimal effect on the bioavailability of oral liquid levothyroxine.
Liothyronine: (Moderate) Proton pump inhibitors (PPIs) may reduce the oral absorption of thyroid hormones and thus reduce efficacy; monitor for altered clinical response to thyroid hormone therapy if concomitant use is necessary. Alternatively, an oral liquid levothyroxine dosage form may be considered. Gastric acidity is an essential requirement for adequate absorption of levothyroxine tablets and capsules and other thyroid hormones. Gastric acidity may be less essential for the absorption of oral liquid dosage forms of levothyroxine; PPIs have been observed to have a minimal effect on the bioavailability of oral liquid levothyroxine.
Lisinopril; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Loop diuretics: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and loop diuretic use due to risk for hypomagnesemia.
Losartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Mefloquine: (Moderate) Proton pump inhibitors (PPIs) may increase plasma concentrations of mefloquine. Patients on chronic mefloquine therapy might be at increased risk of adverse reactions, especially patients with a neurological or psychiatric history.
Methotrexate: (Major) Avoid concomitant use of methotrexate and proton pump inhibitors (PPIs) due to the risk of severe methotrexate-related adverse reactions. If concomitant use is unavoidable, closely monitor for adverse reactions; consider temporary withdrawal of the PPI in some patients receiving high-dose methotrexate. Concomitant use of methotrexate, primarily at high dose, and PPIs may increase and prolong serum concentrations of methotrexate, possibly leading to methotrexate toxicities.
metOLazone: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.

Metoprolol; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Mycophenolate: (Moderate) Concomitant administration of proton pump inhibitors (PPIs) with mycophenolate mofetil (Cellcept) appears to reduce MPA exposure AUC-12h (25.8 +/- 6.4 mg/L x h with omeprazole vs. 33.3 +/- 11.5 mg//L x h without omeprazole); however, the interaction does not appear to exist with mycophenolate sodium delayed-release tablets (Myfortic). Reduced systemic exposure of MPA after mycophenolate mofetil in the presence of a PPI appears to be due to impaired absorption of mycophenolate mofetil which may occur because of incomplete dissolution of mycophenolate mofetil in the stomach at elevated pH. The clinical significance of reduced MPA exposure is unknown; however patients should be evaluated periodically if mycophenolate mofetil is administered with a PPI. Of note, MPA concentrations appear to be reduced in the initial hours after mycophenolate mofetil receipt but increase later in the dosing interval because of enterohepatic recirculation. A second peak in the concentration-time profile of MPA is observed 612 hours after dosing due to enterohepatic recirculation. For example, the 12-hour plasma concentrations of MPA were similar among patients who received mycophenolate mofetil with or without omeprazole. The biphasic plasma concentration-time course of MPA due to extensive enterohepatic circulation hampers therapeutic drug monitoring of MPA. Drug exposure as measured by AUC-12h is the best estimator for the clinical effectiveness of mycophenolate, but measurement of full-dose interval MPA AUC-12h requires collection of multiple samples over a 12-hour period; MPA predose concentrations correlate poorly with MPA AUC-12h. The interaction does not appear to exist with Mycophenolate sodium (Myfortic).
Nelfinavir: (Major) Use of proton pump inhibitors with nelfinavir is not recommended. Coadministration may result in decreased nelfinavir exposure, subtherapeutic antiretroviral activity, and possibility resistant HIV mutations. In one study, concurrent use of nelfinavir with omeprazole resulted in decreased nelfinavir AUC, Cmax, and Cmin by 36%, 37%, and 39%, respectively.
Neratinib: (Major) Avoid concomitant use of neratinib with proton pump inhibitors due to decreased absorption and systemic exposure of neratinib; the solubility of neratinib decreases with increasing pH of the GI tract. Concomitant use with lansoprazole decreased neratinib exposure by 65%.
Nilotinib: (Major) Avoid the concomitant use of nilotinib and proton pump inhibitors (PPIs), as PPIs may cause a reduction in nilotinib bioavailability. Nilotinib displays pH-dependent solubility with decreased solubility at a higher pH. PPIs inhibit gastric acid secretion and elevate the gastric pH. Administration of a single 400-mg nilotinib dose with multiple oral doses of esomeprazole 40 mg/day reduced the nilotinib AUC by 34% in a study in healthy subjects. Increasing the dose is unlikely to compensate for the loss

of nilotinib exposure; additionally, separating the administration of these agents may not eliminate the interaction as PPIs affect the pH of the upper GI tract for an extended period of time.
Nirogacestat: (Major) Avoid concomitant use of nirogacestat and proton pump inhibitors. Concurrent use may impair nirogacestat absorption which may decrease nirogacestat exposure and reduce its efficacy. Antacids may be used with nirogacestat but administration should be separated by at least 2 hours.
Norethindrone Acetate; Ethinyl Estradiol; Ferrous fumarate: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Norethindrone; Ethinyl Estradiol; Ferrous fumarate: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Octreotide: (Moderate) Coadministration of oral octreotide with proton pump inhibitors (PPIs) may require increased doses of octreotide. Coadministration of oral octreotide with drugs that alter the pH of the upper GI tract, including PPIs, may alter the absorption of octreotide and lead to a reduction in bioavailability. This interaction has been documented with esomeprazole and can occur with the other PPIs.
Olmesartan; amLODIPine; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Olmesartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Paltusotine: (Moderate) Avoid the use of proton pump inhibitors (PPIs) in patients already on paltusotine 60 mg. For those on other paltusotine doses, monitor for a decrease in paltusotine efficacy during concomitant use of paltusotine and a PPI and increase the paltusotine dosage as appropriate based on response. PPIs alter gastric pH and interfere with paltusotine absorption which may reduce paltusotine exposure and efficacy. Concomitant use reduced the overall exposure of 20 mg and 60 mg of paltusotine by 21% and 42%, respectively.

PAZOPanib: (Major) Pazopanib displays pH-dependent solubility with decreased solubility at a higher pH. The concomitant use of pazopanib and proton pump inhibitors (PPIs) that elevate the gastric pH may reduce the bioavailability of pazopanib. In a study of patients with solid tumors, the AUC and Cmax of pazopanib were decreased by approximately 40% when coadministered with esomeprazole. If a drug is needed to raise the gastric pH, consider use of a short-acting antacid; separate antacid and pazopanib dosing by several hours.
Pexidartinib: (Major) Avoid coadministration of pexidartinib with pantoprazole as concurrent use may decrease pexidartinib exposure which may result in decreased therapeutic response. As an alternative to a proton pump inhibitor (PPI), use locally-acting antacids or H2-receptor antagonists. Coadministration of another PPI decreased pexidartinib exposure by 50%.
Polyethylene Glycol; Electrolytes; Bisacodyl: (Minor) The concomitant use of bisacodyl oral tablets with drugs that raise gastric pH like proton pump inhibitors can cause the enteric coating of the bisacodyl tablets to dissolve prematurely, leading to possible gastric irritation or dyspepsia. When taking bisacodyl tablets, it is advisable to avoid PPIs within 1 hour before or after the bisacodyl dosage.
Posaconazole: (Major) The concurrent use of posaconazole immediate-release oral suspension and proton pump inhibitors (PPIs) should be avoided, if possible, due to the potential for decreased posaconazole efficacy. If used in combination, closely monitor for breakthrough fungal infections. PPIs increase gastric pH, resulting in decreased posaconazole absorption and lower posaconazole plasma concentrations. When a single 400 mg dose of posaconazole oral suspension was administered with esomeprazole (40 mg PO daily), the mean reductions in Cmax were 46% and the mean reductions in AUC were 32% for posaconazole. The pharmacokinetics of posaconazole delayed-release tablets and oral suspension are not significantly affected by PPIs. Additionally, posaconazole is a potent inhibitor of CYP3A4, an isoenzyme partially responsible for the metabolism of many PPIs (dexlansoprazole, esomeprazole, lansoprazole, omeprazole, pantoprazole, and rabeprazole). Coadministration may result in increased plasma concentration of the PPIs.
Quinapril; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Rilpivirine: (Contraindicated) Concurrent use of proton pump inhibitors and rilpivirine is contraindicated; when these drugs are coadministered, there is a potential for treatment failure and/or the development of rilpivirine or NNRTI resistance. Proton pump inhibitors inhibit secretion of gastric acid by proton pumps thereby increasing the gastric pH; for optimal absorption, rilpivirine requires an acidic environment. Coadministration of a proton pump inhibitor and rilpivirine may result in decreased rilpivirine absorption/serum concentrations, which could cause impaired virologic

response to rilpivirine.
Rilzabrutinib: (Major) Avoid concomitant use of rilzabrutinib and proton pump inhibitors (PPIs). Concurrent use may decrease rilzabrutinib exposure which may reduce its efficacy. Rilzabrutinib is pH soluble; PPIs raise gastric pH and interfere with rilzabrutinib absorption. Concomitant use with a PPI reduced rilzabrutinib overall exposure by 51%.
Risedronate: (Moderate) Use of proton pump inhibitors (PPIs) with delayed-release risedronate tablets (Atelvia) is not recommended. Co-administration of drugs that raise stomach pH increases risedronate bioavailability due to faster release of the drug from the enteric coated tablet. This interaction does not apply to risedronate immediate-release tablets. In healthy subjects who received esomeprazole for 6 days, the Cmax and AUC of a single dose of risedronate delayed-release tablets (Atelvia) increased by 60% and 22%, respectively. PPIsare widely used and are frequently coadministered in users of oral bisphosphonates. A national register-based, open cohort study of 38,088 elderly patients suggests that those who use PPIs in conjunction with alendronate have a dose-dependent loss of protection against hip fracture. While causality was not investigated, the dose-response relationship noted during the study suggested that PPIs may reduce oral alendronate efficacy, perhaps through an effect on absorption or other mechanism, and therefore PPIs may not be optimal agents to control gastrointestinal complaints. Study results suggest that the interaction may occur across the class; however, other interactions have not been confirmed and data suggest that fracture protection is not diminished when risedronate is used with PPIs. A post hoc analysis of patients who took risedronate 5 mg daily during placebo-controlled clinical trials determined that risedronate significantly reduced the risk of new vertebral fractures compared to placebo, regardless of concomitant PPI use. PPI users (n = 240) and PPI non-users (n = 2489) experienced fracture risk reductions of 57% (p = 0.009) and 38% (p < 0.001), respectively.
Saquinavir: (Major) Coadministration with omeprazole results in significantly increased saquinavir concentrations. A similar interaction is expected with all proton pump inhibitors (PPIs). If saquinavir must be administered with PPIs, the patient should be closely monitored for saquinavir-related toxicities, including gastrointestinal symptoms, increased triglycerides, and deep vein thrombosis (DVT). Coadministration with omeprazole results in significantly increased saquinavir concentrations. In a small study, 18 healthy individuals received saquinavir 1000 mg (with ritonavir 100 mg) twice daily for 15 days; on days 11 through 15 omeprazole 40 mg was given once daily, which resulted in an 82% increase in the saquinavir AUC. A similar interaction is expected with all PPIs.
Secretin: (Major) Discontinue proton pump inhibitors (PPIs) before performing a secretin stimulation test for gastrinoma. PPIs may cause a hyperresponse in gastrin secretion and interfere with the test's diagnostic accuracy. The time it takes gastrin concentrations to return to baseline following PPI discontinuation is highly variable.
Selpercatinib: (Major) Avoid coadministration of selpercatinib with pantoprazole due to

the risk of decreased selpercatinib exposure which may reduce its efficacy. If concomitant use is unavoidable, selpercatinib must be taken with food. Coadministration under fasting conditions with another proton pump inhibitor decreased selpercatinib exposure by 69%; however, concomitant use increased selpercatinib exposure by 2% or less when it was administered with a meal.
Sodium Ferric Gluconate Complex; Ferric Pyrophosphate Citrate: (Moderate) The bioavailability of oral iron salts is influenced by gastric pH, and the concomitant administration of proton pump inhibitors can decrease iron absorption. The non-heme ferric form of iron needs an acidic intragastric pH to be reduced to ferrous and to be absorbed. Iron salts and polysaccharide-iron complex provide non-heme iron. Proton pump inhibitors have long-lasting effects on the secretion of gastric acid and thus, increase the pH of the stomach. The increase in intragastric pH can interfere with the absorption of iron salts.
Sofosbuvir; Velpatasvir: (Major) Coadministration of proton pump inhibitors (PPIs) with velpatasvir is not recommended. If it is considered medically necessary to coadminister, velpatasvir should be administered with food and taken 4 hours before omeprazole 20 mg. Other PPIs have not been studied; however, it may be prudent to separate the administration of the other PPIs similarly. Velpatasvir solubility decreases as pH increases; therefore, drugs that increase gastric pH are expected to decrease the concentrations of velpatasvir, potentially resulting in loss of antiviral efficacy.
Sofosbuvir; Velpatasvir; Voxilaprevir: (Major) Coadministration of proton pump inhibitors (PPIs) with velpatasvir is not recommended. If it is considered medically necessary to coadminister, velpatasvir should be administered with food and taken 4 hours before omeprazole 20 mg. Other PPIs have not been studied; however, it may be prudent to separate the administration of the other PPIs similarly. Velpatasvir solubility decreases as pH increases; therefore, drugs that increase gastric pH are expected to decrease the concentrations of velpatasvir, potentially resulting in loss of antiviral efficacy.
Solifenacin: (Moderate) The American College of Gastroenterology states that the effectiveness of proton pump inhibitors (PPIs) may be theoretically decreased if given with other antisecretory agents (e.g., anticholinergics). Proton pump inhibitors (PPIs) inhibit only actively secreting H+-pumps.
Sotorasib: (Major) Avoid coadministration of sotorasib and gastric acid-reducing agents, such as proton pump inhibitors (PPIs). Coadministration may decrease sotorasib exposure resulting in decreased efficacy. If necessary, sotorasib may be administered 4 hours before or 10 hours after a locally acting antacid. Coadministration with a PPI decreased sotorasib exposure by 57% under fed conditions and 42% under fasted conditions.
Sparsentan: (Major) Avoid concurrent use of sparsentan and proton pump inhibitors (PPIs) due to the risk for decreased sparsentan exposure which may reduce its efficacy. Medications that affect gastric pH may reduce sparsentan absorption.

Spironolactone; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
St. John's Wort, Hypericum perforatum: (Major) Avoid concomitant use of St. John's wort with the proton pump inhibitors (PPIs) as PPI exposure may be decreased, reducing their efficacy. PPIs are CYP3A4 and CYP2C19 substrates and St. John's wort is a strong CYP3A4 and CYP2C19 inducer. For example, coadministration of omeprazole with St. John's wort decreased omeprazole plasma concentrations by approximately 40%.
Taletrectinib: (Major) Avoid concomitant use of taletrectinib and proton pump inhibitors (PPIs). Concurrent use may decrease taletrectinib exposure, which may reduce its efficacy. Taletrectinib oral absorption is pH dependent and PPIs alter gastric pH. Concomitant use with a PPI reduced taletrectinib overall exposure by 40%.
Telmisartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Thiazide diuretics: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Thyroid hormones: (Moderate) Proton pump inhibitors (PPIs) may reduce the oral absorption of thyroid hormones and thus reduce efficacy; monitor for altered clinical response to thyroid hormone therapy if concomitant use is necessary. Alternatively, an oral liquid levothyroxine dosage form may be considered. Gastric acidity is an essential requirement for adequate absorption of levothyroxine tablets and capsules and other thyroid hormones. Gastric acidity may be less essential for the absorption of oral liquid dosage forms of levothyroxine; PPIs have been observed to have a minimal effect on the bioavailability of oral liquid levothyroxine.
Tipranavir: (Moderate) Some manufacturers recommend avoiding the coadministration of hepatic cytochrome P-450 enzyme inducers and proton pump inhibitors (PPIs). Tipranavir markedly induces the hepatic cytochrome P-450 enzyme CYP2C19, an enzyme responsible for the metabolism of PPIs. However, since tipranavir is not given unless it is co-prescribed with ritonavir, a known marked enzyme inhibitor, a reduction in PPI metabolism may be unlikely to occur. A reduction in PPI concentrations may increase the risk of gastrointestinal (GI) adverse events such as GI bleeding. If tipranavir and PPIs must be used together, monitor the patient closely for signs and symptoms of GI bleeding or other signs and symptoms of reduced PPI efficacy.
Tolterodine: (Moderate) The American College of Gastroenterology states that the effectiveness of proton pump inhibitors (PPIs) may be theoretically decreased if given with other antisecretory agents (e.g., anticholinergics). Proton pump inhibitors (PPIs) inhibit only actively secreting H+-pumps.
Torsemide: (Moderate) Monitor magnesium concentration before and periodically

during concomitant pantoprazole and loop diuretic use due to risk for hypomagnesemia.
Triamterene; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Trospium: (Moderate) The American College of Gastroenterology states that the effectiveness of proton pump inhibitors (PPIs) may be theoretically decreased if given with other antisecretory agents (e.g., anticholinergics). Proton pump inhibitors (PPIs) inhibit only actively secreting H+-pumps.
Valsartan; hydroCHLOROthiazide, HCTZ: (Moderate) Monitor magnesium concentration before and periodically during concomitant pantoprazole and thiazide diuretic use due to risk for hypomagnesemia.
Warfarin: (Moderate) Monitor the INR in patients receiving warfarin with proton pump inhibitors. Increases in INR may lead to abnormal bleeding. Adjust the warfarin dose to maintain the target INR.
Xanomeline; Trospium: (Moderate) The American College of Gastroenterology states that the effectiveness of proton pump inhibitors (PPIs) may be theoretically decreased if given with other antisecretory agents (e.g., anticholinergics). Proton pump inhibitors (PPIs) inhibit only actively secreting H+-pumps.
Ziftomenib: (Major) Avoid concomitant use of ziftomenib and a proton pump inhibitor (PPI). Consider alternatives, such as an H2-blocker or antacid, if an agent that alters gastric pH is required; adjustments to timing of administration may also be necessary. Concurrent use may interfere with ziftomenib absorption which may reduce ziftomenib exposure and efficacy. Concomitant use of ziftomenib and a PPI reduced ziftomenib overall exposure by 53%.

Adverse Reaction

General Information

Significant elevations in serum gastrin have been reported with pantoprazole similar to other PPIs; this effect may be dose-related. Carcinoid tumors are a potential sequelae of prolonged hypergastrinemia secondary to the use of proton pump inhibitors due to inhibited gastric acid secretion. In 39 patients treated with oral pantoprazole 40 mg to 240 mg daily (majority receiving 40 mg to 80 mg) for up to 5 years, there was a moderate increase in enterochromaffin-like (ECL) cell density starting after the first year of use which appeared to plateau after 4 years. In rodents, pantoprazole is carcinogenic and caused rare types of gastrointestinal tumors. The relevance of these animal findings to humans is unknown. In long-term rodent studies, tumors, including rare types of gastrointestinal tumors, were observed. The relevance of animal findings to human risk is unknown. The safety and efficacy of pantoprazole beyond 12 months have not been

established. The risk of carcinoid tumors during therapy with proton pump inhibitors is low based on cumulative safety experience; monitoring of serum gastrin levels during PPI therapy is generally not necessary.

Some reports of overdosage with pantoprazole have been received. A spontaneous report of a suicide involving an overdosage of pantoprazole (560 mg) has been received; however, the death was more reasonably attributed to the unknown doses of chloroquine and zopiclone which were also taken. Two other reported cases of pantoprazole overdosage involved similar amounts of oral pantoprazole (400 and 600 mg) with no adverse effects observed. One study patient with refractory peptic ulcer disease received 320 mg/day PO for 3 months; the treatment was well tolerated. Doses of up to 240 mg/day IV, have been administered to healthy subjects for 7 days and have been well tolerated.

abdominal pain, constipation, diarrhea, dizziness, dysgeusia, dyspepsia, flatulence, gastric polyps, GI bleeding, headache, nausea, pancreatitis, vomiting, weight gain, weight loss, xerostomia

The safety profile of oral pantoprazole is consistent with that of other proton pump inhibitors, with the most frequently reported adverse events being GI related. In clinical trials of adult patients with GERD, oral pantoprazole (20 mg or 40 mg) was compared to other acid suppressant therapy (H2-blocker or other PPI) as well as placebo. Nausea was reported in adult oral pantoprazole trials (7%), in pediatric trials (4% or less), as well as in IV pantoprazole trials (more than 1%). Abdominal pain was reported in adult oral pantoprazole trials (6.2%), IV pantoprazole studies (more than 1%), and in the pediatric setting (more than 4%). Similarly, vomiting was reported in adult oral pantoprazole trials (4.3%) and among pediatric patients (more than 4%). Diarrhea was reported in 8.8% of adult patients, more than 4% of pediatric patients receiving oral therapy, and more than 1% of pediatric patients receiving IV therapy. In oral pantoprazole trials, flatulence was reported in 3.9% pantoprazole adult patients and in 4% or less pediatric patients. Finally, constipation has been reported across oral and IV pantoprazole evaluations (adult: 2% or less; pediatric: 4% or less; IV: more than 1%). Additional GI related adverse events reported in oral pantoprazole studies include xerostomia (2% or less). Dyspepsia (more than 1%) was reported in clinical trials with intravenous pantoprazole. GI bleeding (less than 1%) has been reported in association with oral pantoprazole. Other commonly reported, non-GI adverse reactions reported in adult oral pantoprazole trials for pantoprazole, a comparator, and placebo were headache (12.2%, 12.8%, and 8.5%, respectively) and dizziness (3%, 2.9%, and 1.2%, respectively). Among pediatric patients and recipients of IV pantoprazole, headache (pediatric: more than 4%; IV: more than 1%) and dizziness (pediatric: 4% or less; IV: more than 1%) were also reported. During postmarketing surveillance of IV pantoprazole, increased salivation and pancreatitis

have been reported. Dysgeusia and ageusia were reported during postmarketing surveillance of both oral and IV pantoprazole. Adverse events reported for patients receiving oral pantoprazole in Zollinger-Ellison Syndrome (80 mg/day to 240 mg/day for up to 2 years) trials were similar to adult GERD trials. Postmarketing surveillance of oral pantoprazole has identified reports of gastric polyps/fundic gland polyps and both weight gain and weight loss, which may have a relationship to drug exposure.

cholecystitis, cholelithiasis, elevated hepatic enzymes, hepatic failure, hepatitis, hyperbilirubinemia, jaundice

In adult oral pantoprazole studies, hepatitis was reported in 2% or less of patients receiving pantoprazole. In both adult and pediatric studies elevated hepatic enzymes (adult: 2% or less; pediatric: 4% or less) have been reported in association with pantoprazole administration. Hyperbilirubinemia, cholecystitis, cholelithiasis have been reported in less than 1% of patients receiving oral pantoprazole. In addition, hepatocellular damage leading to jaundice and hepatic failure have been reported during post-marketing surveillance of oral and IV pantoprazole.

fever, infection, rhinitis

In IV pantoprazole trials, rhinitis was reported in more than 1% of patients receiving pantoprazole. During oral pantoprazole therapy studies, fever was reported in 2 % or less of patients. Upper respiratory tract infections (more than 4%), fever (more than 4%), and otitis media were reported in pediatric patients receiving pantoprazole. In addition, increasing evidence suggests a link between acid-suppression therapy and pneumonia (community- and hospital-acquired). Several mechanisms have been proposed to account for this association. One such mechanism states that gastric pH serves as a barrier against pathogenic colonization of the gastrointestinal tract. An increase in gastric pH allows for bacterial and viral invasion which, in theory, can precipitate respiratory infections. Another proposed mechanism accounts for the role that gastric acid may have on stimulating the cough reflex that allows for the clearing of infectious agents from the respiratory tract. Finally, the fact that acid-suppressive therapy may impair white blood cell function, which in turn may lead to a depressed immune response to an infection, is listed among possible mechanisms. Regardless of the mechanism, the use of H2-blockers and/or PPIs has been associated with the development of pneumonia. Data from a large epidemiological trial, including 364,683 individuals who developed 5551 first occurrences of community-acquired pneumonia (CAP), suggest an increased risk of developing CAP among users of acid-suppressive therapy compared to those who stopped therapy. After adjusting for confounders, the adjusted relative risk (RR) for CAP among PPI users compared to those who stopped therapy was 1.89 (95% CI, 1.36 to 2.62). Likewise, users of H2-blockers had an adjusted

RR of 1.63 (95% CI, 1.07 to 2.48) compared to those who stopped therapy. In a second large cohort trial, including 63,878 hospital admissions, acid-suppressive therapy was ordered in 52% (83% PPI and 23% H2- blocker, with some patients exposed to both) of new admissions. Hospital-acquired pneumonia occurred in 2,219 admissions (3.5%) with a higher incidence recorded among acid-suppressive therapy exposed patients compared to non-exposed patients. A subset analysis found a statistically significant association between PPI use (OR, 1.3; 95% CI, 1.1 to 1.4) and pneumonia. A non-significant association was found with H2-blockers (OR, 1.2; 95% CI, 0.98 to 1.4); however, the lack of significance was attributed to the studies lack of power to detect significance for an OR of less than 1.3. Until more is known about the relationship between acid-suppression and pneumonia, clinicians are encouraged to carefully select patients before empirically initiating acid-suppressive therapy with H2-blockers or PPIs. A causal relationship between the use of pantoprazole and pneumonia has not been established.

agranulocytosis, anemia, ecchymosis, eosinophilia, leukocytosis, leukopenia, pancytopenia, pernicious anemia, thrombocytopenia, vitamin B12 deficiency

During oral pantoprazole clinical studies, hematological reactions that have been reported in less than 2% of patients include leukopenia and thrombocytopenia. Anemia, ecchymosis, eosinophilia, hypochromic anemia, iron deficiency anemia, leukocytosis have also been identified in less than 1% of oral pantoprazole treated patients. Pancytopenia and agranulocytosis have been reported during oral and IV pantoprazole, post-marketing surveillance. Long-term (e.g., generally 2 to 3 years or more) treatment with acid-suppressing agents can lead to malabsorption of vitamin B12 (cyanocobalamin). In a study of healthy volunteers, it was shown that omeprazole caused a significant reduction in cyanocobalamin absorption (vitamin B12 deficiency). One large case-controlled study compared patients with and without an incident diagnosis of vitamin B12 deficiency (n = 25,956 and 184,199, respectively). A correlation was demonstrated between vitamin B12 deficiency and gastric acid-suppression therapy. Patients receiving 2 years or more of a proton pump inhibitor (PPI)(OR, 1.65 [95% CI, 1.58 to 1.73]) or 2 years or more of a H2-receptor antagonist (OR, 1.25 [95% CI, 1.17 to 1.34]) were associated with having an increased risk for vitamin B12 deficiency. A dose-dependant relationship was evident, as daily doses more than 1.5 PPI pills/day were more strongly associated with vitamin B12 deficiency (OR, 1.95 [95% CI, 1.77 to 2.15]) compared to daily doses less than 0.75 pills/day (OR, 1.63 [95% CI, 1.48 to 1.78]; p = 0.007 for interaction). The possibility of cyanocobalamin deficiency and pernicious anemia should be considered if clinical symptoms are observed. Neurological manifestations of pernicious anemia can occur in the absence of hematologic changes.

asthenia, blurred vision, confusion, depression, drowsiness, fatigue, hallucinations, insomnia, malaise, seizures, tinnitus, vertigo

During clinical evaluation of oral pantoprazole in adult patients with GERD, depression and vertigo were reported in 2% or less of pantoprazole treated patients. Vertigo was also reported in 4% or less of pediatric patients (1 to 16 years) receiving oral pantoprazole for the treatment of erosive esophagitis associated with GERD as well as during IV pantoprazole post-marketing surveillance. Asthenia, confusion, drowsiness (somnolence), fatigue, hallucinations, insomnia and malaise were reported during post-marketing surveillance of oral pantoprazole. Seizures were noted in less than 1% of oral pantoprazole treated patients in clinical studies. In IV pantoprazole studies, insomnia was reported in 1% or more of patients. Other nervous system or special senses adverse events reported during IV pantoprazole postmarketing surveillance include anterior ischemic optic neuropathy, blurred vision, hypokinesia, speech disorder, and tinnitus.

acute generalized exanthematous pustulosis (AGEP), anaphylactic shock, anaphylactoid reactions, angioedema, Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS), edema, erythema multiforme, injection site reaction, phlebitis, photosensitivity, pruritus, rash, Stevens-Johnson syndrome, toxic epidermal necrolysis, urticaria

Allergic reactions (adult: 2% or less; pediatric: 4% or less), including urticaria (adult: 2% or less; pediatric: 4% or less) and rash (adult: 2% or less; pediatric: more than 4%), have been reported in patients treated with oral pantoprazole. Other dermatological reactions observed in 2% or less of adult patients treated with oral pantoprazole are photosensitivity reaction and pruritus. Facial edema has been reported in both adult (2% or less) and pediatric (4% or less) studies as well. During postmarketing surveillance of oral and/or IV pantoprazole, the following adverse reactions were identified: anaphylactoid reactions (including anaphylactic shock), angioedema (Quincke's edema), and severe dermatologic reactions (some fatal) including erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (TEN, some fatal), drug reaction with eosinophilia and systemic symptoms (DRESS), and acute generalized exanthematous pustulosis (AGEP). Injection site reaction including thrombo-phlebitis and abscess have been reported in more than 1% of patients treated with IV pantoprazole. Assess the patient and remove the catheter if clinically indicated if thrombophlebitis occurs. Discontinue pantoprazole and consider further evaluation if patient develops signs or symptoms of severe cutaneous adverse reactions or other signs of hypersensitivity.

impotence (erectile dysfunction), interstitial nephritis

Acute tubulo-interstitial nephritis (AIN or TIN) has been observed in patients taking PPIs including with pantoprazole. Acute interstitial nephritis may occur at any point during PPI therapy and is generally attributed to an idiopathic hypersensitivity reaction. Patients may present with varying signs and symptoms from symptomatic hypersensitivity reactions to non-specific symptoms of decreased renal function (e.g., malaise, nausea, anorexia). In reported case series, some patients were diagnosed on biopsy and in the absence of extra-renal manifestations (e.g., fever, rash or arthralgia). Discontinue pantoprazole and evaluate patients with suspected acute AIN. In some cases, corticosteroid treatment and withdrawal of pantoprazole resulted in recovery of renal function. Genitourinary (GU) reactions have also been reported with PPI use. Impotence (erectile dysfunction) has been reported during postmarketing experience with pantoprazole.

lupus-like symptoms

Cutaneous lupus erythematosus (CLE), systemic lupus erythematosus (SLE), and lupus-like symptoms have occurred in patients taking PPIs, including pantoprazole. Both exacerbation and new onset of existing autoimmune disease have be reported, with the majority of PPI-induced lupus erythematosus cases being CLE. Subacute CLE (SCLE) is the most common form of CLE reported in patients treated with PPIs, occurring within weeks to years after continuous drug therapy in patients ranging from infants to geriatric adults. Histological findings were usually observed without organ involvement. SLE is less commonly reported; PPI associated SLE is generally milder than non-drug induced SLE. Onset of SLE typically occurred within days to years after initiating treatment primarily in patients ranging from infants to the elderly. Most patients presented with rash; however, arthralgia and cytopenia were also reported. Do not administer PPIs for longer than medically indicated. If signs or symptoms consistent with CLE or SLE occur, discontinue the drug and refer the patient to the appropriate specialist for evaluation. Most patients improve with discontinuation of the PPI alone in 4 to 12 weeks; serological testing (ANA) may be positive and elevated serological test results may take longer to resolve than clinical manifestations.

arthralgia, bone fractures, elevated creatine kinase, myalgia, rhabdomyolysis

Pantoprazole therapy has been associated with certain musculoskeletal adverse reactions. During clinical studies, arthralgia was reported in 2.8% of adults and 4% or less of pediatric patients (1 to 16 years) receiving oral pantoprazole. Reports of myalgia (adult: 2% or less; pediatric: 4% or less) were identified across all studied age groups. During postmarketing surveillance of oral and/or IV pantoprazole, bone fractures,

elevated creatine kinase (4% or less of patients), and rhabdomyolysis have been reported. Several published observational studies suggest that proton pump inhibitor (PPI) therapy may be associated with an increased risk for osteoporosis-related bone fractures of the hip, wrist, or spine. The risk of fracture was increased in patients who received high-dose (defined as multiple daily doses) and long-term PPI therapy (a year or longer). Use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated. In patients at risk for osteoporosis-related fractures, manage according to established treatment guidelines and ensure adequate vitamin D and calcium supplementation.

hypocalcemia, hypokalemia, hypomagnesemia, hyponatremia

Hypomagnesemia, hypocalcemia, hypokalemia, and hyponatremia have been reported during postmarketing pantoprazole use. Cases of hypomagnesemia have been reported in association with prolonged (3 months to more than 1 year) proton pump inhibitor (PPI) use. Hypomagnesemia may lead to hypocalcemia and/or hypokalemia and may exacerbate underlying hypocalcemia in patients at risk. Low serum magnesium may lead to serious adverse reactions such as muscle spasm (tetany), seizures, and irregular heartbeat (arrhythmias). Consider monitoring electrolyte concentrations and supplementing electrolytes when needed. Discontinuation of PPI therapy may be necessary.

angina, arrhythmia exacerbation, atrial fibrillation, atrial flutter, bleeding, heart failure, hypertension, hypertriglyceridemia, hypotension, migraine, myocardial infarction, palpitations, peripheral vasodilation, sinus tachycardia, syncope, thrombosis

Hypertriglyceridemia (adult: 2% or less; pediatric: 4% or less), has been reported during oral pantoprazole clinical trials. During clinical evaluation of patients with erosive esophagitis associated GERD receiving oral pantoprazole, reported cardiovascular adverse events occurring in less than 1% include: abnormal electrocardiogram, angina pectoris, arrhythmia exacerbation, atrial fibrillation, atrial flutter, cardiovascular disorder, chest pain substernal, congestive heart failure, hemorrhage/bleeding, hypertension, hypotension, migraine, myocardial infarction, palpitations, peripheral vasodilation, retinal vascular disorder, syncope, sinus tachycardia, and thrombosis. In most instances, the relationship to pantoprazole therapy was unclear.

C. difficile-associated diarrhea, pseudomembranous colitis, superinfection

C. difficile-associated diarrhea (CDAD) or pseudomembranous colitis has been reported with pantoprazole. PPI therapy, such as pantoprazole, may be associated with an

increased risk of CDAD, especially in hospitalized patients. The use of gastric acid suppressive therapy, such as PPIs, may increase the risk of enteric infection or superinfection by encouraging the growth of gut microflora. If pseudomembranous colitis is suspected or confirmed, institute appropriate fluid and electrolyte management, protein supplementation, C. difficile-directed antibacterial therapy, and surgical evaluation as clinically appropriate.

laboratory test interference

Administration of pantoprazole may result in laboratory test interference, specifically serum chromogranin A (CgA) tests for neuroendocrine tumors, urine tests for tetrahydrocannabinol (THC), secretin stimulation tests, and diagnostic tests for H. pylori. Gastric acid suppression may increase serum CgA. Increased CgA concentrations may cause false positive results in diagnostic investigations for neuroendocrine tumors. To prevent this interference, temporarily stop pantoprazole at least 14 days before assessing CgA concentrations and consider repeating the test if initial concentrations are high. If serial tests are performed, ensure the same commercial laboratory is used as reference ranges may vary. Reports have suggested use of proton pump inhibitors (PPIs) may cause false positive urine screening tests for THC. If a PPI-induced false positive urine screen is suspected, confirm the positive results using an alternative testing method. Pantoprazole may cause a hyper-response in gastrin secretion to the secretin stimulation test, falsely suggesting gastrinoma. Health care providers are advised to temporarily stop pantoprazole at least 14 days prior to performing a secretin stimulation test to allow gastrin concentrations to return to baseline. PPIs are known to suppress H. pylori; thus, ingestion of a PPI within 2 weeks of performing diagnostic tests for H. pylori may lead to false negative results. At a minimum, instruct the patient to avoid the use of pantoprazole for at least 2 weeks prior to the test.

gastric polyps

Proton pump inhibitor (PPI) use is associated with an increased risk of fundic gland gastric polyps. The risk for gastric polyps increases with long-term PPI use, especially beyond 1 year. Gastric polyps/fundic gland polyps have been reported during postmarketing surveillance with pantoprazole. Patients are usually asymptomatic, and the polyps are identified incidentally on endoscopy. Gastroduodenal carcinoids have been reported in patients with Zollinger-Ellison (ZE) syndrome receiving long-term pantoprazole; however, the ZE condition itself is known to be associated with such tumors. The overall risk of carcinoid tumors during PPI therapy is low based on cumulative safety experience; monitoring of serum gastrin levels during PPI therapy is generally not necessary. In one trial of patients (n = 25) with H2-receptor antagonist-

resistant gastroesophageal reflux disease (GERD) treated with long-term (4 years or more) pantoprazole therapy, no cases of neoplasia or dysplasia were seen in biopsies. Use the shortest duration of PPI therapy appropriate to treat the specific condition. Symptomatic response to therapy with pantoprazole does not preclude the presence of gastric cancer or other malignancy. Consider additional follow-up and diagnostic testing in adults who have a suboptimal response or an early symptomatic relapse after completing treatment with a PPI. In older adult patients, also consider an endoscopy.

drug-induced hypergastrinemia, gastric hypersecretion

Studies suggest that long-term PPI therapy is associated with temporal gastric hypersecretion shortly following treatment discontinuation. A similar and well-established response has been noted after withdrawal of H2 blockers. Profound gastric acid suppression during PPI therapy leads to a drug-induced hypergastrinemia and subsequent rebound acid hypersecretion. In this hypersecretory state, enterochromaffin-like cell hypertrophy also results in a temporal increase in serum chromogranin A (CgA) levels. It is unclear, however, if this hypersecretory reflex results in clinically significant effects in patients on or attempting to discontinue PPI therapy. A clinically significant effect may lead to gastric acid-related symptoms upon PPI withdrawal and possible therapy dependence. Studies in healthy subjects (H. pylori negative) as well as GERD patients, present conflicting data regarding whether PPI therapy beyond 8-weeks is associated with rebound acid hypersecretion and associated dyspeptic symptoms shortly following PPI cessation. Until more consistent study results shed light on this possible effect, it is prudent to follow current treatment guidelines employing the lowest effective dose, for the shortest duration of time in symptomatic patients. For patients requiring maintenance therapy, consider on demand or intermittent PPI therapy, step down therapy to an H2 blocker, and regularly assess the need for continued gastric suppressive therapy.

Description

Pantoprazole is an oral and intravenous gastric proton pump inhibitor (PPI). It is indicated in adults for the short-term treatment and maintenance therapy of erosive esophagitis associated with gastroesophageal reflux disease (GERD) and for the treatment of Zollinger-Ellison syndrome. In pediatric patients, pantoprazole is primarily used for GERD and erosive esophagitis. Other uses for pantoprazole include stress ulcer prophylaxis and the treatment of peptic ulcer disease. It has been shown to be effective in the management of duodenal ulcer or gastric ulcer, including those caused by Helicobacter pylori (H. pylori). Pantoprazole, in combination with appropriate antibiotic therapy, is used to eradicate H. pylori. When compared to omeprazole (20 mg once daily), pantoprazole (40 mg once daily) has more potent antisecretory activity, but

similar clinical efficacy in treating peptic ulcer disease or GERD. On the basis of available research, no clear advantage has been demonstrated for the use of 1 PPI over another in the treatment of GERD. There are, however, wide variations in the acid-suppression potency with the following ranked from least to highest potency: pantoprazole, lansoprazole, omeprazole, esomeprazole, and rabeprazole. Therefore, a 1-time switch to a different PPI in a refractory patient may be useful. A key to optimizing effectiveness is tailoring dosage timing and considering twice-daily dosing; traditional delayed-release PPIs should be administered 30 to 60 minutes before a meal for maximal pH control. PPIs are often used for stress ulcer prophylaxis in critically ill patients; however, evidence for benefit is lacking, and use may increase the risk of adverse reactions such as pneumonia and Clostridioides difficile infection. Reserve stress ulcer prophylaxis for patients with risk factors for clinically significant gastrointestinal bleeding (e.g., multiple organ dysfunction, prolonged mechanical ventilation, coagulopathy, persistent shock, concomitant corticosteroid therapy). Pantoprazole does not significantly inhibit hepatic CYP isoenzymes and compared to other currently available PPIs, it has the lowest potential for hepatic drug interactions. Pantoprazole was FDA-approved in 2000.

Mechanism Of Action

Pantoprazole is a substituted benzimidazole proton-pump inhibitor (PPI) that suppresses gastric acid secretion by inhibiting the gastric (H+,K+)-ATPase enzyme pump. Pantoprazole forms a covalent bond to 2 sites of the (H+,K+)-ATPase enzyme system at the secretory surface of the gastric parietal cell; this binding results in antisecretory effects that persists more than 24 hours, which allows for once-daily dosing. A significant increase in gastric pH and decrease in basal acid output follow oral and IV administration of pantoprazole. Pantoprazole does not antagonize H2 or cholinergic receptors.

Most studies report that gastric acid suppression is dose-related following IV and oral administration of pantoprazole, resulting in inhibition of both basal and stimulated gastric acid secretion (irrespective of the stimulus). In a series of basal gastric acid secretion studies, at oral doses ranging from 20 to 120 mg, pantoprazole resulted in dose-related increases in the median basal gastric pH and in the percent of time the gastric pH exceeded 3 and 4. Treatment with 40 mg pantoprazole produced optimal increases in gastric pH which were significantly higher than gastric pH following the 20 mg dose. Doses higher than 40 mg (60, 80, 120 mg) generally did not result in further significant increases in median gastric pH. Under stimulated gastric acid secretion using pentagastrin, a dose-dependent decrease in gastric acid output occurs following single oral (20 to 80 mg) or IV (20 to 120 mg) doses of pantoprazole in healthy volunteers. Following the initial oral dose of 40 mg, a 51% mean inhibition is achieved in about 2.5 hours. With once daily dosing (7 days), the mean inhibition increases to 85%. Acid

secretion returns to normal within 7 days of discontinuing pantoprazole, without evidence of rebound hypersecretion.

Significant in vitro activity against Helicobacter pylori (H. pylori) has been demonstrated for pantoprazole. In one report, the minimum inhibitory concentration (MIC90) for pantoprazole (100 mcg/mL) was higher than that observed for omeprazole (25 mcg/mL) and lansoprazole (6.25 mcg/mL); however, the clinical significance of this finding is unknown.

Similar to omeprazole and other PPIs, hypergastrinemia can occur during pantoprazole therapy. Although prolonged hypergastrinemia has been associated with gastric tumors in rats, long-term studies of proton pump inhibitors do not suggest the development of tumors in humans.

No clinically relevant effects of pantoprazole on cardiovascular, respiratory, ophthalmic, or central nervous system function have been detected. A short-term (2-week) evaluation showed that oral pantoprazole had no effect on endocrine hormones including: cortisol, testosterone, triiodothyronine (T3), thyroxine (T4), thyroid-stimulating hormone, thyronine-binding protein, parathyroid hormone, insulin, glucagon, renin, aldosterone, follicle-stimulating hormone, luteinizing hormone, prolactin and growth hormone.

Pharmacokinetics

Pantoprazole is administered orally or intravenously. It is 99% bound to human plasma proteins, primarily to albumin. Pantoprazole is extensively metabolized in the liver to inactive metabolites by the cytochrome P450 (CYP) system. The primary metabolic pathway is demethylation by CYP2C19, with subsequent sulfation; a minor metabolic pathway is oxidation by CYP3A4. The terminal elimination half-life following IV or oral administration is approximately 1 hour. Poor metabolizers have longer elimination half-life values ranging from 3.5 to 10 hours; however, they have minimal drug accumulation (23% or less) with once daily dosing. About 71% of the dose is excreted renally and 18% is eliminated by biliary excretion. No unchanged pantoprazole is recovered in the urine.

Affected cytochrome P450 isoenzymes and drug transporters: CYP2C19, CYP2C9, CYP3A4
Pantoprazole is extensively metabolized in the liver to inactive metabolites via CYP2C19 (primary pathway) and CYP3A4 (minor pathway). In vitro data suggest that pantoprazole is a competitive inhibitor of CYP2C9 and CYP3A4 isoenzymes; however, the effects do not appear to result in clinically relevant drug interactions. In vitro and clinical data suggest it is a weak inhibitor of CYP2C19 isoenzymes.

Route-Specific Pharmacokinetics

- **Oral Route**

Protonix delayed-release tablets are enteric-coated to allow pantoprazole sodium sesquihydrate, which is acid labile, to pass though the stomach intact. After oral administration of delayed-release tablets, peak plasma concentrations occur in about 2.5 hours. Pantoprazole is well absorbed; it undergoes little first-pass metabolism resulting in an absolute bioavailability of approximately 77%. Food delays, but does not affect the extent of absorption of the tablets; tablets may be administered without regard to meals. However, the delayed-release oral suspension should be administered approximately 30 minutes before a meal.

- **Intravenous Route**

The onset of antisecretory effects of pantoprazole following IV administration is approximately 15 to 30 minutes, with a duration of action of 24 hours.

- **Hepatic Impairment**

Pantoprazole serum concentrations are increased in people with hepatic disease. In patients with mild to moderate hepatic impairment, Cmax increased 1.5-fold, elimination half-life values increased to 7 to 9 hours, and the AUC values increased 5 to 7-fold relative to healthy subjects; however, minimal drug accumulation occurred during once daily dosing.

- **Renal Impairment**

Pantoprazole serum concentrations are not affected by renal dysfunction. In patients with severe renal impairment, the pharmacokinetic parameters for pantoprazole were similar to those of healthy subjects. Pantoprazole is not removed by hemodialysis.

- **Pediatrics**

Neonates and Infants

The pharmacokinetics of pantoprazole have been found to be highly variable in pediatric patients; although the exposure after oral doses in neonates and infants appears similar to that seen after oral doses in adults, the predicted clearance has been shown to be faster in younger pediatric patients compared to adults. One study analyzed the results of either a 0.6 mg/kg PO dose or a 1.2 mg/kg PO dose of oral granules in infants aged 1 to less than 12 months (n = 42) as compared with exposures obtained in adults at 40-mg/day dosing. Exposure increased with increasing doses of pantoprazole granules, even though wide interindividual variability was observed. Compared with that in adults receiving pantoprazole 40 mg, exposure obtained with the 1.2 mg/kg dose was similar to that obtained with adult usual dosing. The average half-life was 1.78 +/- 1.3 hours. In a trial of 20 patients (age, 10 days to 16.4 years), 156 pantoprazole concentration measurements were used to analyze the pharmacokinetics of pantoprazole doses ranging from 19.9 to 140.6 mg/1.73 m2/day. Data showed that pantoprazole clearance

increased with age and weight and for typical patients aged 6 months to 5 years, the predicted pantoprazole clearance is faster than that reported in adults.

Children and Adolescents
The pharmacokinetics of pantoprazole have been found to be highly variable in pediatric patients; the predicted clearance has been shown to be faster in younger pediatric patients compared to adults. One study analyzed the results of either a 0.6 mg/kg PO dose or a 1.2 mg/kg PO dose of oral granules in children aged 1 year to less than 6 years (n = 17) as compared with exposures obtained in adults at 40-mg/day dosing. Exposure increased with increasing doses of pantoprazole granules, even though wide interindividual variability was observed. Compared with that in adults receiving pantoprazole 40 mg, exposure obtained with the 1.2 mg/kg dose was slightly lower to that obtained with adult usual dosing. The average half-life was 1.7 +/- 0.6 hours. In another trial of 20 patients (age, 10 days to 16.4 years), 156 pantoprazole concentration measurements were used to analyze the pharmacokinetics of pantoprazole doses ranging from 19.9 to 140.6 mg/1.73 m2/day. Data showed that pantoprazole clearance increased with age and weight and for typical patients aged 6 months to 5 years, the predicted pantoprazole clearance is faster than that reported in adults. In a population pharmacokinetic analysis, total clearance increased in a non-linear fashion as bodyweight increased. In a separate analysis, the pharmacokinetics of pantoprazole tablets were evaluated in pediatric patients 6 to 16 years of age. After a single 40 mg dose, the mean estimated AUC was about 39% higher in 6 to 11 year old (n = 12) and 10% higher in 12 to 16 year old (n = 11) patients, compared to AUC values in adults. The median time to peak plasma concentration (Tmax) was 2 hours.

- **Geriatric**

Pantoprazole serum concentrations are increased in the geriatric adult. In older versus younger adult subjects, pantoprazole AUC and Cmax values were increased by 43% and 26% respectively following multiple oral doses; however, no dosage adjustment is recommended based on adult age. After repeated intravenous administration in older adult subjects (65 to 76 years of age), the AUC and elimination half-life values of pantoprazole were similar to those observed in younger subjects.

- **Gender Differences**

Pantoprazole serum concentrations are not affected by gender.

- **Ethnic Differences**

The primary metabolic pathway of pantoprazole is demethylation by CYP2C19, with subsequent sulfation; a minor metabolic pathway is oxidation by CYP3A4 isoenzymes. CYP2C19 displays a known genetic polymorphism due to its deficiency in some sub-populations (e.g., 3% of Caucasians and African-Americans and 17% to 23% of Asians are poor metabolizers of CYP2C19). Poor metabolizers of pantoprazole have longer

elimination half-life values ranging from 3.5 to 10 hours; however, they have minimal drug accumulation (23% or less) with once daily dosing.

Administration

For storage information, see the specific product information within the How Supplied section.

Oral Administration

Oral Solid Formulations

Delayed-release tablets
The manufacturer states that pantoprazole may be administered without regard to meals; however, guidelines recommend pantoprazole be administered on an empty stomach 30 to 60 minutes before meals. If given once daily, administer before the first meal of the day. Administer twice daily doses before the first and dinner meals to optimize response.
Swallow whole. Do not chew, crush, or split. If the patient is unable to swallow a 40-mg tablet, then two 20-mg tablets may be prescribed.

Oral Liquid Formulations

Delayed-release oral suspension granules
Do not divide the suspension packet to obtain a smaller dosage.
Administer approximately 30 minutes before a meal.
Do not crush or chew the granules.
May sprinkle on 5 mL (1 teaspoonful) of applesauce; administer within 10 minutes. Alternatively, mix in a small cup with 5 mL (1 teaspoonful) of apple juice; stir for 5 seconds. The granules will not break up. Administer right away, then rinse the container with additional apple juice and administer, to assure delivery of the full dose. Do not prepare using any other liquids or foods, including water.

Nasogastric (NG) tube administration
Empty the contents of 1 packet of oral suspension granules into the barrel of a 60 mL (2 ounce) catheter tip syringe (with the plunger removed) that is connected to a 16 French (or larger) nasogastric tube. Add 10 mL of apple juice and gently tap and/or shake the barrel of the syringe to help empty the syringe into the tube. Repeat at least twice more using the same amount of apple juice (10 mL or 2 teaspoonfuls) each time. No granules should remain in the syringe.

Extemporaneous Compounding-Oral

Extemporaneous 2 mg/mL pantoprazole oral suspension preparation
Count out 20 of the Protonix 40-mg oral tablets.
Remove the Protonix imprint from all the tablets by gently rubbing the tablets on a paper towel dampened with ethanol. Let the tablets air dry.
Crush and triturate the tablets in a mortar and reduce to a coarse powder.
Transfer crushed tablets to a 600 mL beaker, and add 340 mL sterile water for irrigation, USP.
Place beaker on a magnetic stirrer.
While stirring, add 16.8 g of sodium bicarbonate powder and stir for about 20 minutes until the tablet remnants have disintegrated and coating has dissolved.
While stirring, add another 16.8 g of sodium bicarbonate powder and stir for about 5 minutes until powder has dissolved.
Add enough sterile water for irrigation, USP to bring the final volume to 400 mL. Mix well.
Final concentration of suspension: pantoprazole 2 mg/mL.
Transfer to amber-colored bottle.
Shake well prior to each use.
For accurate dosing, measure dosage using an oral syringe or other calibrated device.
Storage: Suspension is stable for 62 days when stored under refrigeration at 35.6 to 46.4 degrees F (2 to 8 degrees C) and protected from light. Label with "shake well".

Injectable Administration

For intravenous use only.
Visually inspect parenteral products for particulate matter and discoloration prior to administration whenever solution and container permit.

Intravenous Administration

General Information
Administer pantoprazole IV either through a dedicated IV line or a Y-site. When administered via a Y-site, immediately stop use if precipitation or discoloration occurs. Pantoprazole is incompatible with midazolam during Y-site coadministration and pantoprazole may be incompatible with IV solutions containing zinc salts. Pantoprazole injections contain edetate disodium(the salt form of EDTA), a chelator of metal ions including zinc. Caution should be used when other EDTA containing products are also co-administered intravenously.
Do not use spiked IV system adapters with Protonix IV vials as glass vial breakage may occur.

Pantoprazole powder for solution for injection
IV Push preparation and administration (2-minute infusion)

Dilute each 40 mg vial with 10 mL 0.9% Sodium Chloride Injection. Each vial will have a final concentration of 4 mg/mL.
Flush the IV line with 5% Dextrose Injection or 0.9% Sodium Chloride Injection before and after each dose.
Infuse slowly over at least 2 minutes.
Do not administer with other IV fluids or medications.
Storage: Store for up to 24 hours at room temperature prior to administration. Reconstituted vials do not require protection from light. Do not freeze. One study concluded that the 4 mg/mL reconstituted solution is stable in polypropylene syringes for at least 96 hours when stored under refrigeration (3 to 5 degrees C) or at room temperature (23 to 25 degrees C). To minimize the potential for solution discoloration, refrigerate if stored beyond 48 hours.
Intermittent IV Infusion preparation and administration (15-minute infusion)
Erosive Esophagitis (Erosive GERD)
Reconstitute each 40 mg vial with 10 mL 0.9% Sodium Chloride Injection.
Dilute the reconstituted vial
Pediatric patients 3 months to 11 months: Dilute with 21 mL of 0.9% Sodium Chloride Injection to a final concentration of approximately 1.3 mg/mL.
Pediatric patients 1 year to 17 years old and adult patients: Further dilute with 100 mL 5% Dextrose Injection or 0.9% Sodium Chloride Injection to a final concentration of approximately 0.4 mg/mL.
Withdraw the desired dose of the diluted dose for the pediatric or adult dose.
Flush the IV line with 5% Dextrose Injection or 0.9% Sodium Chloride Injection before and after each dose.
Infuse over approximately 15 minutes.
Do not administer by IV push or other parenteral routes other than IV infusion (i.e., intramuscular administration, subcutaneous administration).
Do not administer with other IV fluids or medications.
Storage: The reconstituted solution may be stored up to 6 hours at room temperature (up to 30 degrees C or 86 degrees F) prior to further dilution. Store the diluted solution at room temperature and use within 24 hours from the time of initial reconstitution. Do not freeze the reconstituted or diluted solution.
Hypersecretory Syndromes
Reconstitute two (2) 40 mg vials with 10 mL of 0.9% Sodium Chloride Injection.
Combine the contents of the 2 vials and dilute with 80 mL of 5% Dextrose injection or sodium chloride injection to a total volume of 100 mL with a final concentration of approximately 0.8 mg/mL.
Flush the IV line with 5% Dextrose Injection and 0.9% Sodium Chloride Injection before and after each dose.
Administer IV over a period of approximately 15 minutes at a rate of approximately 7

mL/minute.
Storage: Store the reconstituted solution at room temperature for up to 24 hours from the time of initial reconstitution. Do not freeze.
Continuous Intravenous Infusion
NOTE: Pantoprazole is not FDA-approved by the FDA to be administered by continuous IV infusion, but is under FDA review. Several adult consensus guidelines recommend a pantoprazole 80 mg IV infusion bolus, followed by 8 mg/hour continuous infusion in the treatment of selected indications.
Dilute each 40-mg vial with 10 mL 0.9% Sodium Chloride Injection. Each vial will have a final concentration of 4 mg/mL.
Prior to infusion, further admix 2 reconstituted vials (80 mg/20 mL total) with 80 mL of 0.9% Sodium Chloride Injection, to a final concentration of approximately 0.8 mg/mL.
ASHP Recommended Standard Concentration for Pediatric Continuous Infusions: 0.8 mg/mL.
Give the 80 mg IV bolus over a period of 30 minutes followed by a continuous infusion of 8 mg/hour.
Storage: Store at room temperature and use within 24 hours from the time of initial reconstitution.

Pantoprazole solution for injection
Preparation
Thaw frozen container at room temperature 20 to 25 degrees C (68 to 77 degrees F) or under refrigeration 2 to 8 degrees C (36 to 46 degrees F). Products should not be thawed by immersion in water baths or by microwave irradiation. Do not force thaw.
No further dilution is necessary.
Check for minute leaks by squeezing the container firmly. If leaks are detected, discard solution as sterility may be impaired.
Do not add supplementary medication.
Inspect thawed pantoprazole solution visually for particulate matter prior to and during administration. Thawed solution is expected to range from colorless to yellow over time.
Storage: Thawed solution is stable for 21 days under refrigeration 2 to 8 degrees C (36 to 46 degrees F) or 48 hours at room temperature 20 to 25 degrees C (68 to 77 degrees F). Protect from light when stored under refrigeration. Do not refreeze the thawed solution.
IV Administration
Administer pantoprazole IV either through a dedicated IV line or a Y-site. When administered via a Y-site, immediately stop use if precipitation or discoloration occurs. Flush the IV line with 5% Dextrose Injection and 0.9% Sodium Chloride Injection before and after each dose.
When administered through a Y-site, pantoprazole is compatible with 5% Dextrose Injection and 0.9% Sodium Chloride Injection. Pantoprazole is incompatible with

midazolam during Y-site coadministration and pantoprazole may be incompatible with IV solutions containing zinc salts.
Infuse over approximately 15 minutes.
Do not administer with other IV fluids or medications.

Maximum Dosage Limits

- **Adults**
  80 mg/day PO/IV for most indications; up to 160 mg/day PO has been used off-label for H. pylori eradication; up to 240 mg/day PO/IV for Zollinger-Ellison syndrome.
- **Geriatric**
  80 mg/day PO/IV for most indications; up to 160 mg/day PO has been used off-label for H. pylori eradication; up to 240 mg/day PO/IV for Zollinger-Ellison syndrome.
- **Adolescents**
  weighing more than 40 kg: 40 mg/day PO is FDA-approved maximum; however, up to 80 mg/day PO has been used off-label for H. pylori eradication; 40 mg/day IV.
  weighing 40 kg: 40 mg/day PO is FDA-approved maximum; however, up to 80 mg/day PO has been used off-label for H. pylori eradication; 20 mg/day IV.
  weighing 16 kg to 39 kg: 20 mg/day PO is FDA-approved maximum; however, up to 2.5 mg/kg/day (Max: 80 mg/day) PO has been used off-label for H. pylori eradication; 20 mg/day IV.
  weighing 15 kg or less: 20 mg/day PO is FDA-approved maximum; however, up to 2.5 mg/kg/day (Max: 80 mg/day) PO has been used off-label for H. pylori eradication; 10 mg/day IV.
- **Children**
  5 to 12 years weighing more than 40 kg: 40 mg/day PO is FDA-approved maximum; however, up to 80 mg/day PO has been used off-label for H. pylori eradication; 40 mg/day IV.
  5 to 12 years weighing 40 kg: 40 mg/day PO is FDA-approved maximum; however, up to 80 mg/day PO has been used off-label for H. pylori eradication; 20 mg/day IV.
  5 to 12 years weighing 16 to 39 kg: 20 mg/day PO is FDA-approved maximum; however, up to 2.5 mg/kg/day (Max: 80 mg/day) PO has been used off-label for H. pylori eradication; 20 mg/day IV.
  5 to 12 years weighing 15 kg: 20 mg/day PO is FDA-approved maximum; however, up to 2.5 mg/kg/day (Max: 80 mg/day) PO has been used off-label for H. pylori eradication; 10 mg/day IV.
  5 to 12 years weighing less than 15 kg: up to 2.5 mg/kg/day PO has been used off-label; 10 mg/day IV.
  1 to 4 years weighing more than 40 kg: up to 2.5 mg/kg/day PO has been used off-

label; 40 mg/day IV.
1 to 4 years weighing 16 to 40 kg: up to 2.5 mg/kg/day PO has been used off-label; 20 mg/day IV.
1 to 4 years weighing 15 kg or less: up to 2.5 mg/kg/day PO has been used off-label; 10 mg/day IV.

- **Infants**
  3 to 11 months weighing 12.5 kg or more: 10 mg/day IV; 1.2 mg/kg/day PO off-label.
  3 to 11 months weighing less than 12.5 kg: 0.8 mg/kg/day IV; 1.2 mg/kg/day PO off-label.
  1 to 2 months: limited data available for off-label IV use; 1.2 mg/kg/day PO off-label.
- **Neonates**
  1.2 mg/kg/day PO suggested by limited off-label data; very limited data available for off-label IV use.

Dosage Forms

- Pantoprazole 0.4mg/1mL Solution for injection
- Pantoprazole 0.8mg/1mL Solution for injection
- Pantoprazole Sodium 20mg Oral tablet, gastro-resistant
- Pantoprazole Sodium 40mg Granules for oral suspension
- Pantoprazole Sodium 40mg Oral tablet, gastro-resistant
- Pantoprazole Sodium 40mg Powder for solution for injection
- Pantoprazole Sodium Bulk powder
- Protonix 20mg Delayed-Release Tablet
- Protonix 40mg Delayed-Release Granules for Suspension
- Protonix 40mg Delayed-Release Tablet
- Protonix 40mg Powder for Injection
- Protonix 40mg Powder for Injection (PREMIER ProRx)

Dosage Adjustment Guidelines

Hepatic Impairment

No dosage adjustment is necessary.

Renal Impairment

No dosage adjustment is necessary.

Intermittent hemodialysis
No dosage adjustment is necessary. Pantoprazole is not removed by hemodialysis.

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